RESOLVING ERDŐS–ULAM MONOCHROMATIC UNION-CLOSED FAMILY CONJECTURES

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Abstract. We prove both Erdős–Ulam conjectures on monochromatic union-closed families. Every two-colouring of the subsets of a finite set contains a monochromatic union-closed family whose size grows faster than any fixed power of the size of the set, while suitable colourings admit no monochromatic union-closed family of exponential size. Both results hold for any number of colours. For sublattices the forced size exceeds the chain bound and is determined up to a logarithmic factor. Every theorem is checked in Lean.

1. Introduction

Write $[n] = \{1, \dots, n\}$ and $2^{[n]}$ for the family of all subsets of $[n]$. For an integer $k \geq 2$, a *k-colouring* of $2^{[n]}$ is a map $\chi : 2^{[n]} \to [k]$, and a family $\mathcal{F} \subseteq 2^{[n]}$ is *monochromatic* if χ is constant on $\mathcal{F}$. The family $\mathcal{F}$ is *union-closed* if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$, and it is a *sublattice* if it is closed under both union and intersection. Erdős and Ulam asked how large a monochromatic family of either kind a colouring is forced to contain. Accordingly let

$$f(n) = \min_{\chi} \max\big\{|\mathcal{L}| : \ \mathcal{L} \subseteq 2^{[n]} \text{ is a monochromatic sublattice}\big\},$$

$$F_k(n) = \min_{\chi} \max\big\{|\mathcal{F}| : \ \mathcal{F} \subseteq 2^{[n]} \text{ is a monochromatic union-closed family}\big\},$$

where the first minimum is over all 2-colourings and the second over all k-colourings of $2^{[n]}$, and write $F = F_2$. A single set is a sublattice and every sublattice is union-closed, so $1 \leq f(n) \leq F(n)$. A maximal chain $\emptyset \subset \{1\} \subset \{1,2\} \subset \cdots \subset [n]$ has $n+1$ members, at least $\lceil (n+1)/2 \rceil$ of them receive the same colour under a 2-colouring, and every subchain is a sublattice. Hence

$$F(n) \ \geq \ f(n) \ \geq \ \left\lceil \frac{n+1}{2} \right\rceil. \tag{1.1}$$

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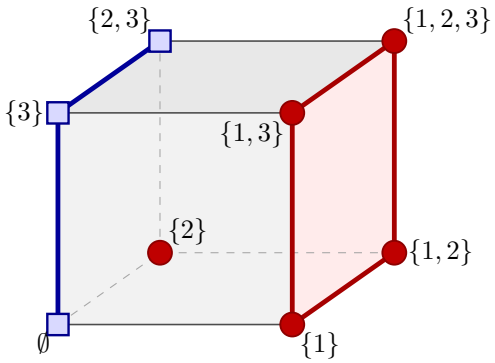


FIGURE 1. A 2-colouring of the subsets of $\{1,2,3\}$, drawn on the cube $2^{[3]}$. The five red sets (discs) form a union-closed family which is not a sublattice, since $\{1\} \cap \{2\} = \emptyset$ is blue. The three blue sets (squares) form a chain.

A 2-colouring is also a k-colouring, so $F_k(n) \le F(n)$ for every $k \ge 2$. Figure 1 shows the two kinds of family for $n = 3$.

The question is Problem 1183 in Bloom's collection of Erdős problems [Bl26]. The problem asks for estimates of $f(n)$ and $F(n)$. As recorded there, Erdős had no plausible conjecture for the true order of magnitude of $f(n)$ and no good guess for that of $F(n)$, but asked in particular whether

$$F(n) \;\ge\; n^{\omega(n)} \;\text{ for some } \omega(n) \to \infty, \qquad \text{and} \qquad F(n) \;<\; (1+o(1))^n. \tag{1.2}$$

Erdős also noted that Howorka had proved the first statement for colourings in which the colour of a set depends only on its size.[1] We prove both statements in (1.2) for arbitrary colourings, and for any number of colours. The second statement was proved earlier by Chojecki [Ch26], in the form $F(n) \le n^{\log_2 n + O(\log\log n)}$, together with $f(n) \le (3+o(1))\, n \log_2 n$; his lower bounds are the chain bound (1.1).

1.1. **Results.** The main result is a polynomial lower bound of every degree, with an explicit constant.

Theorem 1.1. *Let $k \ge 2$ and $t \ge 1$ be integers, put $M = t^2 k^{2^t}$ and*

$$c_{k,t} = \left(k^{2^{t+1}}\, t^t\, (4M^2)^{t+1}\right)^{-1}.$$

Every k-colouring of $2^{[n]}$ contains a monochromatic union-closed family with at least $c_{k,t}\, n^{t+1}$ members; that is, $F_k(n) \ge c_{k,t}\, n^{t+1}$ for all $n \ge 1$. For two colours, $F(n) \ge n^t$ whenever $\log_2 n \ge (t+2)^2\, 2^t$.

[1] The problem page states Howorka's result without a reference, and we have not found Howorka's argument in print. Section 6 gives a complete proof, with an explicit bound in terms of van der Waerden numbers.

The upper bound comes from a random colouring and the Sauer–Shelah lemma, as in Chojecki's proof of $F(n) \le n^{\log_2 n + O(\log\log n)}$ [Ch26].

Theorem 1.2. *Let $d \ge 1$ be an integer with $2^d > nd + 2$. Then*

$$F(n) \;\le\; \sum_{j=0}^{d-1} \binom{n}{j}.$$

In particular, for $n \ge 2$ and $d = \lfloor \log_2 n \rfloor + \lfloor \log_2 \lfloor \log_2 n \rfloor \rfloor + 3$,

$$F(n) \;\le\; d\, n^{d-1} \;\le\; \bigl(\log_2 n + \log_2\log_2 n + 3\bigr)\, n^{\log_2 n + \log_2\log_2 n + 2}.$$

Together the two theorems settle both parts of (1.2), and they locate the exponent: the largest $\omega(n)$ with $F(n) \ge n^{\omega(n)}$ lies between $(1-o(1))\log_2\log_2 n$ and $(1+o(1))\log_2 n$.

Corollary 1.3 (The Erdős–Ulam conjectures)**.** *Both statements in (1.2) hold, in the following form.*

(i) *For every $n \ge 2$ we have $F(n) \ge n^{\omega(n)}$, where*

$$\omega(n) = \ell - 2\log_2(\ell+2) - 1, \qquad \ell = \log_2\log_2 n,$$

and $\omega(n) \to \infty$.

(ii) *There is a function $\varepsilon : \mathbb{N} \to \mathbb{R}$ with $\varepsilon(n) \to 0$ such that $F(n) < (1+\varepsilon(n))^n$ for every $n \ge 1$.*

(iii) *If $\omega : \mathbb{N} \to \mathbb{R}$ satisfies $F(n) \ge n^{\omega(n)}$ for every n, then for every $\varepsilon > 0$ we have $\omega(n) \le (1+\varepsilon)\log_2 n$ for all large n.*

(iv) *For every $k \ge 2$ there are functions $\omega_k(n) \to \infty$ and $\varepsilon_k(n) \to 0$ such that $n^{\omega_k(n)} \le F_k(n) < (1+\varepsilon_k(n))^n$ for every $n \ge 1$.*

Parts (i) and (ii) are the two statements of (1.2), with the term $o(1)$ written out as a function $\varepsilon(n) \to 0$, so Corollary 1.3 answers both of these questions in the affirmative. The estimates of $f(n)$ and $F(n)$ that the problem asks for in the first place are the bounds of Theorems 1.1, 1.2 and 1.4, and their exact orders remain open (Section 9). Since $\log_2(\ell+2) = o(\ell)$, the exponent in (i) is $(1-o(1))\log_2\log_2 n$. This is also, to first order, the size of the monochromatic union-closed families that we can exhibit in a uniformly random colouring (Proposition 5.2), so a better lower bound for $F(n)$ would have to say something new about random colourings as well. In the other direction, Proposition 5.4 reduces an upper bound for the random colouring to counting union-closed families, and Proposition 5.5 shows that this count is too large to bound $F(n)$ below $2^{n/4}$.

The two functions of Erdős and Ulam have very different sizes. For sublattices the trivial bound (1.1) is correct up to a factor of order $\log n$, but for large n it is not sharp.

Theorem 1.4. *For every $n \geq 1$, with $q = \lfloor (n+1)/12 \rfloor$, $L = \lfloor \log_2 n \rfloor + 1$ and $K = \lfloor \log_2 L \rfloor + 1$,*

$$\left\lceil \frac{n+1+q}{2} \right\rceil \;\leq\; f(n) \;\leq\; \min\bigl\{\, n^2+n+1,\;\; 2n(L+K+3)+L+K+4 \,\bigr\}.$$

In particular $f(n) > \lceil (n+1)/2 \rceil$ for every odd $n \geq 11$ and every $n \geq 23$, and $13n/24 \leq f(n) \leq (2+o(1))\, n \log_2 n$, whereas $F(n)/n^C \to \infty$ for every constant C.

The lower bound comes from chains in which stretches of twelve consecutive levels are replaced by small sublattices. In each stretch the red and the blue sets of the new family together outnumber the twelve sets of the chain: either two sets of the same size have different colours and both can be used (Lemma 7.2), or the colour of a set depends only on its size, and then a short combinatorial argument on words in two letters finds a product of chains that beats the chain in one colour (Lemmas 7.3 and 7.4). The upper bound rests on a count of sublattices by size: deleting the points one at a time shows that $2^{[n]}$ has at most $(m+1)^{2n}$ nonempty sublattices with at most m members (Lemma 7.8). Chojecki [Ch26] had obtained $f(n) \leq (3+o(1))\, n \log_2 n$ by counting posets of bounded width.

1.2. **Colourings by cardinality, Hilbert cubes and progressions.** Howorka's case of the problem is arithmetic in nature. Call χ a *cardinality colouring* if $\chi(S) = g(|S|)$ for some $g : \{0,1,\dots,n\} \to [k]$. The building blocks of our families are the *t-cubes* $\{U \cup X_I :\; I \subseteq [t]\}$, where $U, X_1, \dots, X_t$ are pairwise disjoint, the X_i are nonempty and $X_I = \bigcup_{i \in I} X_i$. The sizes of the members of such a cube form the *Hilbert cube*

$$H(u; a_1, \dots, a_t) = \Bigl\{ u + \sum_{i \in I} a_i :\; I \subseteq [t] \Bigr\}, \qquad u = |U|,\; a_i = |X_i|,$$

so a cube is monochromatic under a cardinality colouring exactly when g is constant on the corresponding Hilbert cube. Hilbert's lemma [Hi1892], proved in the course of Hilbert's work on the irreducibility of polynomials with integer coefficients and usually regarded as the first theorem of Ramsey theory, says that every finite colouring of the positive integers has monochromatic Hilbert cubes of every dimension. Our argument gives it with an explicit window (Corollary 6.1): every k-colouring of $\{r, r+1, \dots, r+t^2k^{2^t}\}$ is constant on a t-dimensional Hilbert cube inside this interval. Read through the dictionary between cubes of sets and cubes of integers, the proof of Theorem 1.1 produces many monochromatic Hilbert cubes spread along a single sequence of bounded steps.

Theorem 1.5. *Let $k \geq 2$ and $t \geq 1$, put $M = t^2k^{2^t}$, let $n \geq 4M(t+1)$, and put $m = \lfloor (n-1)/M \rfloor$ and $s = \lfloor m/2 \rfloor$. For every $g : \{0,1,\dots,n\} \to [k]$ there*

are integers $\beta_1, \dots, \beta_m \in [M]$ *with* $\beta_1 + \dots + \beta_m < n$ *and a colour* c *with the following property. For at least*

$$\frac{n^{t+1}}{k\, t^t\, (4M^2)^{t+1}}$$

of the pairs (j, A) *with* $s < j \le m$, $A \subseteq [s]$ *and* $|A| = t$, *the function* g *takes the value* c *at every point of the Hilbert cube*

$$H_\beta(j, A) = \Big\{ \beta_1 + \beta_2 + \dots + \beta_j - \sum_{b \in B} \beta_b : \ B \subseteq A \Big\}.$$

Different pairs (j, A) may give the same set of integers, and there are at most $(n+1)M^t$ Hilbert cubes $H(u; a_1, \dots, a_t)$ in $\{0, \dots, n\}$ with all $a_i \le M$. What Theorem 1.5 provides is one sequence β for which a fixed proportion of the cubes indexed by the pairs (j, A) have a common colour, all of them hanging from the partial sums $\beta_1 + \dots + \beta_j$. This is the structure that lets the corresponding families of sets be glued into one union-closed family. Section 6 also shows that a monochromatic arithmetic progression of sizes yields a large monochromatic union-closed family and a monochromatic sublattice, and derives Howorka's theorem from van der Waerden's theorem with an explicit bound (Theorem 6.4).

1.3. **The idea of the proof of Theorem 1.1.** Chains and cubes are monochromatic union-closed families that are easy to find, but a chain has at most $n+1$ members and a single cube is too rigid to be forced to be large. The families we build are unions of many cubes that fit together. Take pairwise disjoint nonempty blocks $Y_1, \dots, Y_m$, put $C_j = Y_1 \cup \dots \cup Y_j$, and for $j > s$ and $A \subseteq [s]$ consider the cube $Q(j, A) = \{C_j \setminus Y_B : \ B \subseteq A\}$. The union of a member of $Q(j, A)$ and a member of $Q(j', A')$ with $j \le j'$ is again a member of $Q(j', A')$, so the union of any collection of cubes $Q(j, A)$ that are monochromatic in a common colour is union-closed. There are about m^{t+1} pairs (j, A) with $|A| = t$, and it is therefore enough to make a fixed proportion of these cubes monochromatic.

For a fixed chain of blocks this cannot be arranged, since the colouring may have been chosen against it. We therefore randomise twice: the ground set is permuted by a uniformly random σ, and each block size is chosen at random from $\{1, \dots, M\}$. After a random permutation, the probability that a cube is monochromatic depends only on the sizes of its parts (Lemma 2.2). The central step is a supersaturation statement: for every r, some shape whose base size lies between r and $r + M$ is monochromatic with probability at least k^{-2^t} (Proposition 3.2). It follows from an inequality of Cauchy–Schwarz type for products over the vertices of a box (Lemma 3.1), applied to the sets obtained by running along t disjoint chains of elements (Figure 6). The shape of $Q(j, A)$ is governed by $t + 1$ block sizes, those of Y_j

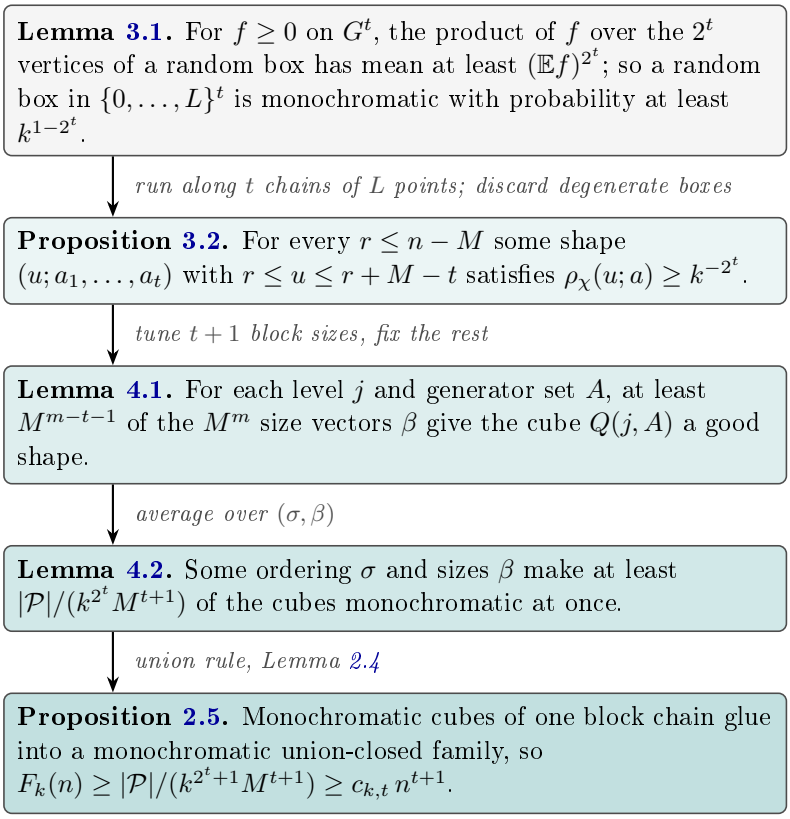


FIGURE 2. The structure of the proof of Theorem 1.1. Here $\mathcal{P}$ is the set of pairs (j, A) with $s < j \leq m$, $A \subseteq [s]$ and $|A| = t$, and ρ_χ is defined after Lemma 2.2.

and of Y_a for $a \in A$, and these can be tuned independently of all the others (Lemma 4.1). Averaging over σ and the block sizes then produces one chain of blocks in which a proportion $k^{-2^t} M^{-t-1}$ of all the cubes is monochromatic (Lemma 4.2). Figure 2 shows how the pieces are assembled.

1.4. **Related work.** The union-closed families in this paper should not be confused with those of Frankl's union-closed sets conjecture, which asks whether every union-closed family other than $\{\emptyset\}$ has an element lying in at least half of its members. That problem concerns the internal structure of one family and involves no colouring. Gilmer [Gi22] proved that some element lies in a constant proportion of the members, and Alweiss, Huang and Sellke [AHS24] improved the constant. The entropy method behind these results compares the union of two independent random members of a given family with a single random member. Here the family is not given in advance but depends on the colouring, and the method does not apply.

Families of 2^d sets of the form $\{U \cup X_I\}$ are called d-dimensional Boolean algebras by Gunderson, Rödl and Sidorenko [GRS99], who studied their extremal theory alongside that of d-dimensional affine cubes of integers and complete d-partite d-uniform hypergraphs. The passage between cubes of sets and Hilbert cubes of integers used in Section 6 is of the same kind. Proposition 3.2 contains the two-letter case of the multidimensional Hales–Jewett theorem [HJ63] with an explicit bound (Remark 3.3); see [GRS90] for this theorem and for the background in Ramsey theory, including van der Waerden's theorem [vdW27]. The box inequality behind it is the simplest instance of the repeated use of the Cauchy–Schwarz inequality introduced by Gowers [Go01]. The upper bound rests on the Sauer–Shelah lemma [Sa72, Sh72], which is also at the root of the theory of Vapnik–Chervonenkis dimension [VC71].

1.5. **Formal verification.** Every result of this paper is proved in the text, and no step of any proof depends on a computer calculation. In addition, the theorems, propositions, corollaries and lemmas of Sections 2–7 have been formalised and checked in the Lean proof assistant [MU21] with the Mathlib library [Mat20], with the exception of Lemmas 7.3 and 7.4. The functions f, F and F_k are defined in the formal development literally as above, the two conjectures are stated in the form (1.2) with the term $o(1)$ written out as a function tending to zero, and Lean reports that the proof of each formal statement uses only the three standard axioms of the system. The formal proofs follow the text, with one exception: for colourings that depend only on the size of a set, the formal proof of Lemma 7.1 replaces the argument of those two lemmas by a finite check carried out inside the logic of the proof assistant. Section 8 describes the formalisation.

1.6. **Organisation.** Section 2 introduces cubes and block chains and proves that monochromatic cubes of one chain can be glued. Section 3 proves the box inequality and the supersaturation statement, and Section 4 completes the proof of Theorem 1.1. Section 5 proves Theorem 1.2 and Corollary 1.3 and studies the random colouring. Section 6 treats cardinality colourings, Hilbert cubes and arithmetic progressions, and Section 7 the lattice function f. Section 8 describes the formalisation, and Section 9 collects open questions.

2. Cubes and block chains

Throughout, $k \geq 2$ and χ is a k-colouring of $2^{[n]}$, and S_n is the group of permutations of $[n]$. A permutation acts on sets and on families of sets in the obvious way: $\sigma(S) = \{\sigma(x) : x \in S\}$ and $\sigma(\mathcal{F}) = \{\sigma(S) : S \in \mathcal{F}\}$. For sets $X_1, X_2, \dots$ and a finite set I of indices we write $X_I = \bigcup_{i \in I} X_i$, so that $X_\emptyset = \emptyset$.

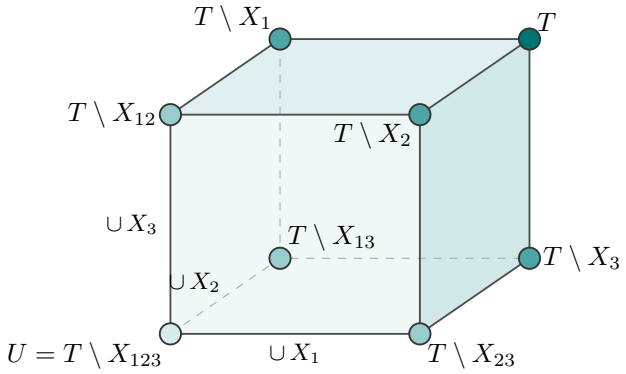


FIGURE 3. A 3-cube with top T and generators X_1, X_2, X_3, where $X_{12} = X_1 \cup X_2$ and so on. Moving along an edge in direction i adds or removes X_i. The vertex $U = T \setminus X_{123}$ is the base.

Definition 2.1. Let $t \geq 0$. A *t-cube* in $2^{[n]}$ is a family

$$Q(U; X_1, \ldots, X_t) = \{U \cup X_I : \ I \subseteq [t]\},$$

where $U, X_1, \ldots, X_t \subseteq [n]$ are pairwise disjoint and $X_1, \ldots, X_t$ are nonempty. We call U the *base*, $T = U \cup X_{[t]}$ the *top* and $X_1, \ldots, X_t$ the *generators*. The *shape* of the cube is the tuple $(|U|; |X_1|, \ldots, |X_t|)$.

The 2^t sets $U \cup X_I$ are distinct, and

$$(U \cup X_I) \cup (U \cup X_J) = U \cup X_{I \cup J}, \qquad (U \cup X_I) \cap (U \cup X_J) = U \cup X_{I \cap J},$$

so a t-cube is a sublattice of $2^{[n]}$ isomorphic to $2^{[t]}$. Writing $U \cup X_I = T \setminus X_{[t] \setminus I}$ describes the same cube from the top downwards, as the family $\{T \setminus X_I : \ I \subseteq [t]\}$; this is the form in which cubes appear in block chains. Figure 3 shows a 3-cube in this form.

Lemma 2.2. *If Q and Q' are t-cubes in $2^{[n]}$ of the same shape, then*

$$\big|\{\sigma \in S_n : \ \sigma(Q) \text{ is monochromatic}\}\big| = \big|\{\sigma \in S_n : \ \sigma(Q') \text{ is monochromatic}\}\big|.$$

Proof. Let $Q = Q(U; X_1, \ldots, X_t)$ and $Q' = Q(U'; X'_1, \ldots, X'_t)$ with tops T and T'. The sets $U, X_1, \ldots, X_t, [n] \setminus T$ partition $[n]$, as do $U', X'_1, \ldots, X'_t, [n] \setminus T'$, and corresponding parts have the same size. So there is $\tau \in S_n$ with $\tau(U) = U'$ and $\tau(X_i) = X'_i$ for every i, and then $\tau(Q) = Q'$. Now $\sigma(Q') = (\sigma\tau)(Q)$, and $\sigma \mapsto \sigma\tau$ is a bijection of S_n. □

For integers $u \geq 0$ and $a_1, \ldots, a_t \geq 1$ with $u + a_1 + \cdots + a_t \leq n$ we may therefore define

$$\rho_\chi(u; a_1, \ldots, a_t) = \frac{1}{n!} \big|\{\sigma \in S_n : \ \sigma(Q) \text{ is monochromatic}\}\big|,$$

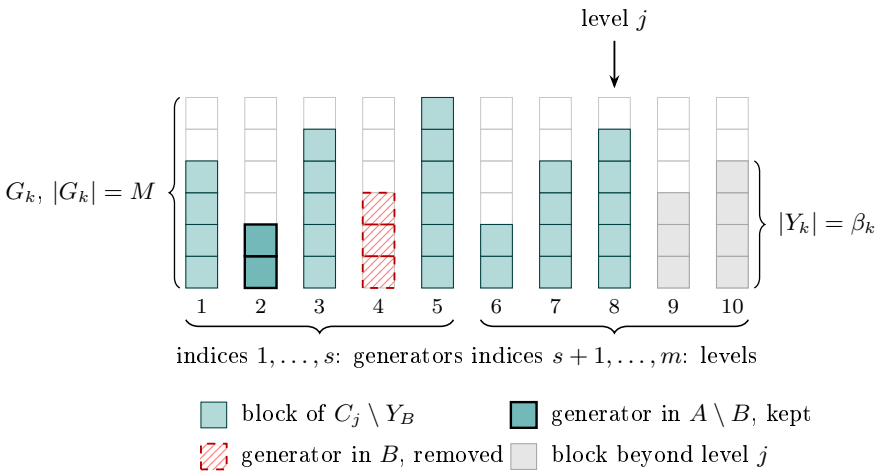


FIGURE 4. A block chain with $m = 10$ blocks placed in slots $G_1, \dots, G_{10}$ of size $M = 6$, with $|Y_k| = \beta_k$ and $\beta = (4, 2, 5, 3, 6, 2, 4, 5, 3, 4)$. Here $s = 5$, and the teal blocks form the member $C_8 \setminus Y_{\{4\}}$ of the cube $Q(8, \{2, 4\})$; the hatched block Y_4 is removed and the grey blocks lie beyond C_8.

where Q is any t-cube of shape $(u; a_1, \dots, a_t)$. Since $\sigma(Q)$ is a uniformly random cube of that shape when σ is uniform, $\rho_\chi(u; a)$ is the probability that a random cube of shape $(u; a)$ is monochromatic.

Definition 2.3. Let $Y_1, \dots, Y_m \subseteq [n]$ be pairwise disjoint nonempty sets, the *blocks*, and put $C_j = Y_1 \cup \cdots \cup Y_j$ for $0 \le j \le m$. For $1 \le j \le m$ and $A \subseteq [j-1]$ let

$$Q(j, A) = \{C_j \setminus Y_B : \ B \subseteq A\}.$$

This is the $|A|$-cube with top C_j, generators Y_a $(a \in A)$ and base $C_j \setminus Y_A$; its shape is $\big(\sum_{i \le j,\, i \notin A} |Y_i|;\ (|Y_a|)_{a \in A}\big)$, the generators being listed in increasing order of a.

Figure 4 shows a member of such a cube. The chain structure enters through the following rule, illustrated in Figure 5.

Lemma 2.4 (Union rule)**.** *Let $1 \le j \le j' \le m$ and $B, B' \subseteq [j]$. Then*

$$(C_j \setminus Y_B) \cup (C_{j'} \setminus Y_{B'}) = C_{j'} \setminus Y_{B \cap B'}.$$

Proof. Both sides are contained in $C_{j'}$. Let $x \in C_{j'}$. If $x \notin Y_{B'}$, then x lies in $C_{j'} \setminus Y_{B'}$ and also outside $Y_{B \cap B'}$. Otherwise $x \in Y_b$ for some $b \in B'$; then $b \le j$, so $x \in C_j$, and x lies in the left side if and only if $b \notin B$, that is, if and only if $x \notin Y_{B \cap B'}$. □

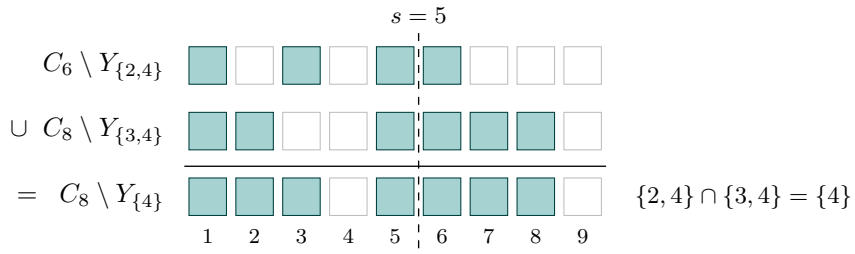


FIGURE 5. The union rule with $s = 5$: $C_6 \setminus Y_{\{2,4\}}$ and $C_8 \setminus Y_{\{3,4\}}$ have union $C_8 \setminus Y_{\{4\}}$. Filled squares are blocks present in the set.

Proposition 2.5 (Gluing). *Let $1 \le s < m$, and let $\mathcal{S}$ be a set of pairs (j, A) with $s < j \le m$ and $A \subseteq [s]$ such that the cube $Q(j, A)$ is monochromatic for every $(j, A) \in \mathcal{S}$. Then there is a monochromatic union-closed family with at least $|\mathcal{S}|/k$ members.*

Proof. Choose a colour c such that the set $\mathcal{S}_c$ of pairs in $\mathcal{S}$ whose cube has colour c satisfies $|\mathcal{S}_c| \ge |\mathcal{S}|/k$, and let $\mathcal{F}$ be the union of the cubes $Q(j, A)$ over $(j, A) \in \mathcal{S}_c$. Every member of $\mathcal{F}$ has colour c.

Let $C_j \setminus Y_B \in Q(j, A)$ and $C_{j'} \setminus Y_{B'} \in Q(j', A')$ be members of $\mathcal{F}$, where $(j, A), (j', A') \in \mathcal{S}_c$ and, say, $j \le j'$. As $B, B' \subseteq [s] \subseteq [j]$, Lemma 2.4 shows that their union is $C_{j'} \setminus Y_{B \cap B'}$, and since $B \cap B' \subseteq A'$ this set belongs to $Q(j', A') \subseteq \mathcal{F}$. So $\mathcal{F}$ is union-closed.

Finally, the map $(j, A) \mapsto C_j \setminus Y_A$ from $\mathcal{S}_c$ to $\mathcal{F}$ is injective. Indeed, $j > s$ means $j \notin A$, so $C_j \setminus Y_A$ meets Y_j and no block of larger index, which determines j; and then A is the set of indices $i \le s$ for which Y_i is disjoint from $C_j \setminus Y_A$. Hence $|\mathcal{F}| \ge |\mathcal{S}_c| \ge |\mathcal{S}|/k$. □

The family $\mathcal{F}$ is not closed under intersection in general: the intersection of the two sets above is $C_j \setminus Y_{B \cup B'}$, and $B \cup B'$ need not be contained in A. By Theorem 1.4 this cannot be repaired: some colourings have no monochromatic sublattice with more than $O(n \log n)$ members.

3. A box inequality and supersaturation

The proof of Theorem 1.1 needs, for every r, a shape whose base size lies a little above r and which is monochromatic with probability bounded below by a quantity depending on k and t alone. This section obtains such a shape from an inequality of Cauchy–Schwarz type for products taken over the vertices of a box.

Let G be a finite set and $t \ge 0$. For $x, x' \in G^t$ and $\omega \in \{0, 1\}^t$ let $x^{(\omega)} \in G^t$ be the point whose i-th coordinate is x'_i if $\omega_i = 1$ and x_i if $\omega_i = 0$. The 2^t points $x^{(\omega)}$ are the vertices of the *box* spanned by x and x'; they are distinct

precisely when $x_i \neq x'_i$ for every i. Expectations are taken with respect to the uniform distribution on the finite set indicated by the variables.

Lemma 3.1 (Box inequality)**.** *Let $f : G^t \to [0,\infty)$. If x and x' are independent and uniformly distributed in G^t, then*

$$\mathbb{E} \prod_{\omega\in\{0,1\}^t} f\big(x^{(\omega)}\big) \;\geq\; \big(\mathbb{E} f(x)\big)^{2^t}.$$

Proof. We use induction on t. For $t = 0$ both sides equal the value of f at the only point of G^0. Let $t \geq 1$ and write the points of G^t as (a, y) with $a \in G$ and $y \in G^{t-1}$. For $y, y' \in G^{t-1}$ and $a \in G$ put

$$h_{y,y'}(a) = \prod_{\omega\in\{0,1\}^{t-1}} f\big(a, y^{(\omega)}\big).$$

Splitting $\omega \in \{0,1\}^t$ according to its first coordinate shows that the product over the box spanned by $x = (a, y)$ and $x' = (a', y')$ equals $h_{y,y'}(a)\, h_{y,y'}(a')$. The variables a, a', y, y' are independent, so the Cauchy–Schwarz inequality gives

$$\mathbb{E} \prod_{\omega} f\big(x^{(\omega)}\big) = \mathbb{E}_{y,y'}\big(\mathbb{E}_a h_{y,y'}(a)\big)^2 \;\geq\; \big(\mathbb{E}_a \mathbb{E}_{y,y'} h_{y,y'}(a)\big)^2.$$

For fixed a, the induction hypothesis applied to $f(a,\cdot)$ gives $\mathbb{E}_{y,y'} h_{y,y'}(a) \geq (\mathbb{E}_y f(a,y))^{2^{t-1}}$, and since $z \mapsto z^{2^{t-1}}$ is convex on $[0,\infty)$, Jensen's inequality gives

$$\mathbb{E}_a\big(\mathbb{E}_y f(a,y)\big)^{2^{t-1}} \;\geq\; \big(\mathbb{E}_a \mathbb{E}_y f(a,y)\big)^{2^{t-1}} = \big(\mathbb{E} f\big)^{2^{t-1}}.$$

Squaring, the left-hand side of the lemma is at least $(\mathbb{E} f)^{2^t}$. □

The quantity on the left of Lemma 3.1 is the 2^t-th power of the box norm of f, and the argument is the one Gowers used for the uniformity norms in a proof of Szemerédi's theorem [Go01].[2] Applied to the indicator of each colour class it says that a random box is monochromatic with probability at least k^{1-2^t}, whatever the colouring. What remains is to turn boxes in a grid into cubes of sets, and this is where the order of the ground set is randomised.

For the rest of the paper put

$$L_k(t) = t\,k^{2^t} \qquad \text{and} \qquad M_k(t) = t\,L_k(t) = t^2\,k^{2^t} \qquad (t \geq 1), \tag{3.1}$$

so that $M_2(1) = 4$, $M_2(2) = 64$ and $M_2(3) = 2304$. Note that $M_k(t) \geq t+1$.

[2] In that setting one works with a function on an abelian group and the vertices of a parallelepiped $x + \omega \cdot h$, and repeated use of the Cauchy–Schwarz inequality compares the average over parallelepipeds with the average of the function. Here the underlying set is a product and the box is a combinatorial one, which is the simplest case of the method.

Proposition 3.2 (Supersaturation)**.** *Let $t \geq 1$ and $L \geq 1$ be integers, put $M = tL$, and let $r \geq 0$ be an integer with $r + M \leq n$. There are integers u and $a_1, \dots, a_t \geq 1$ with*

$$r \leq u, \qquad (u-r)+a_1+\cdots+a_t \leq M, \qquad \rho_\chi(u; a_1, \dots, a_t) \;\geq\; k^{1-2^t} - \frac{t}{L+1}.$$

In particular, with $L = L_k(t)$ and $M = M_k(t)$ as in (3.1), one can achieve $\rho_\chi(u; a_1, \dots, a_t) \geq k^{-2^t}$, and then $u \leq r + M - t$ and $a_i \leq M$ for every i.

Proof. Choose pairwise disjoint sets $R, Z_1, \dots, Z_t \subseteq [n]$ with $|R| = r$ and $|Z_i| = L$, which is possible because $r + tL \leq n$, and list $Z_i = \{z_{i,1}, \dots, z_{i,L}\}$. Put $G = \{0, 1, \dots, L\}$ and, for $x \in G^t$,

$$S(x) = R \cup \bigcup_{i=1}^{t} \{z_{i,q} : \; 1 \leq q \leq x_i\},$$

so that x records how far along each of the t chains $z_{i,1}, z_{i,2}, \dots$ the set $S(x)$ reaches. Let $\sigma \in S_n$ and $x, x' \in G^t$ be independent and uniform, and let E be the event that $\chi(\sigma(S(\cdot)))$ is constant on the box spanned by x and x'.

Fix σ, and for each colour c let p_c be the probability that $\chi(\sigma(S(x))) = c$ for uniform x. Lemma 3.1 applied to the indicator function of the colour class of c shows that, conditionally on σ, all vertices of the box have colour c with probability at least $p_c^{2^t}$. Since $p_1 + \cdots + p_k = 1$, convexity gives $\sum_c p_c^{2^t} \geq k \cdot k^{-2^t}$, and averaging over σ gives $\Pr(E) \geq k^{1-2^t}$. The event D that $x_i = x_i'$ for some i has probability at most $t/(L+1)$, so

$$\Pr(E \setminus D) \;\geq\; k^{1-2^t} - \frac{t}{L+1}.$$

Suppose now that $x_i \neq x_i'$ for every i, and put $\ell_i = \min(x_i, x_i')$ and $h_i = \max(x_i, x_i')$. The vertices of the box are the points whose i-th coordinate is ℓ_i or h_i for every i, so the sets $S(x^{(\omega)})$ form the t-cube $Q_{x,x'} = Q(U; X_1, \dots, X_t)$ with

$$U = S(\ell), \qquad X_i = \{z_{i,q} : \; \ell_i < q \leq h_i\}$$

(Figure 6). Its shape is $(r + \sum_i \ell_i;\; h_1 - \ell_1, \dots, h_t - \ell_t)$, and $u = r + \sum_i \ell_i$ and $a_i = h_i - \ell_i$ satisfy $u \geq r$, $a_i \geq 1$ and $(u-r)+\sum_i a_i = \sum_i h_i \leq tL = M$. The event $E \setminus D$ says that $\sigma(Q_{x,x'})$ is monochromatic. For a fixed pair (x, x') outside D, the probability of this over σ is ρ_χ at the shape of $Q_{x,x'}$, because $\sigma(Q_{x,x'})$ is then a uniformly random cube of that shape. Hence $\Pr(E \setminus D)$ is an average, over all pairs (x, x'), of numbers that vanish on D and are values of ρ_χ at admissible shapes elsewhere, and one of these values is at least $\Pr(E \setminus D)$.

For $L = L_k(t)$ we have $t/(L+1) < t/L = k^{-2^t}$, so the value found is at least $(k-1)k^{-2^t} \geq k^{-2^t}$. The bounds on u and a_i follow from $a_i \geq 1$. $\square$

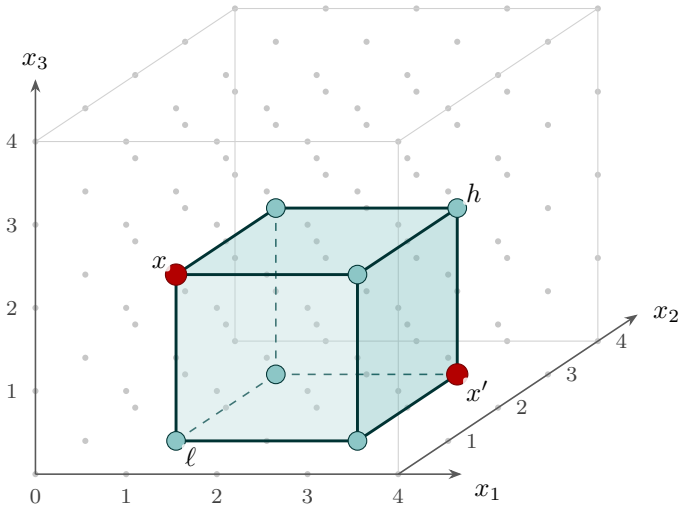


FIGURE 6. The grid G^3 with $G = \{0, 1, \ldots, 4\}$, as in the proof of Proposition 3.2 with $t = 3$ and $L = 4$, and the box spanned by $x = (1, 1, 2)$ and $x' = (3, 3, 0)$. Its eight vertices are the points of $\{1, 3\} \times \{1, 3\} \times \{0, 2\}$, between the corners $\ell = (1, 1, 0)$ and $h = (3, 3, 2)$. The sets $S(y)$ at these vertices form the 3-cube with base $S(\ell)$ and generators $X_i = \{z_{i,q} : \ell_i < q \le h_i\}$, whose shape is $(r + 2; 2, 2, 2)$.

The proof is short enough that its two sources of loss are easy to see.[3] The factor k^{1-2^t} is already the probability that a fixed t-cube is monochromatic in a uniformly random k-colouring, so for such colourings the expected value of ρ_χ at every shape equals k^{1-2^t} and nothing better can be obtained on average. The subtraction of $t/(L + 1)$ only accounts for degenerate boxes, and it is what forces the window M to be as large as $t^2k^{2^t}$.

Remark 3.3. Identify a subset S of a finite set Z with the word in $\{0, 1\}^Z$ that has the letter 1 exactly at the positions in S. A t-cube then becomes a t-dimensional combinatorial subspace: the positions in U carry the fixed letter 1, those outside T the fixed letter 0, and those in X_i the i-th variable. Since $\rho_\chi > 0$ in Proposition 3.2, every k-colouring of the subsets of a set of size $M_k(t)$ is constant on some t-cube, which is the two-letter case of the multidimensional Hales–Jewett theorem [HJ63, GRS90] with the bound $t^2k^{2^t}$. The proposition says more: a proportion at least k^{-2^t} of the cubes

[3]The random ordering σ is what turns a single good shape into a positive proportion of all cubes of that shape. Without it the box inequality would still produce a monochromatic cube in every window, but it would say nothing about how many cubes of the same shape are monochromatic, and it is this density that the averaging of Section 4 consumes.

of one shape are monochromatic, and this density is what makes the rate in Theorem 1.1 explicit.

4. Proof of Theorem 1.1

Fix $k \geq 2$ and $t \geq 1$, put $M = M_k(t)$ as in (3.1), and let n be large enough that

$$m = \Big\lfloor \frac{n-1}{M} \Big\rfloor \qquad \text{and} \qquad s = \Big\lfloor \frac{m}{2} \Big\rfloor$$

satisfy $s \geq t$. Note that $mM \leq n-1$. Fix pairwise disjoint sets $G_1, \dots, G_m \subseteq [n]$ of size M, the *slots*, and list each slot as $G_i = \{g_{i,1}, \dots, g_{i,M}\}$; for instance $g_{i,q} = (i-1)M + q$. For a vector $\beta \in [M]^m$ of block sizes let

$$Y_i^{\beta} = \{g_{i,1}, \dots, g_{i,\beta_i}\} \qquad (1 \leq i \leq m),$$

and for $\sigma \in S_n$ let $Q_{\sigma,\beta}(j, A)$ denote the cube $Q(j, A)$ of Definition 2.3 built from the blocks $\sigma(Y_1^{\beta}), \dots, \sigma(Y_m^{\beta})$. Thus $Q_{\sigma,\beta}(j, A) = \sigma\big(Q_{\mathrm{id},\beta}(j, A)\big)$, and $Q_{\sigma,\beta}(j, A)$ has shape

$$\Big(\sum_{i \leq j,\ i \notin A} \beta_i;\ (\beta_a)_{a \in A} \Big). \tag{4.1}$$

Let $\mathcal{P}$ be the set of pairs (j, A) with $s < j \leq m$, $A \subseteq [s]$ and $|A| = t$. Call β *good for* (j, A) if the shape (4.1) satisfies $\rho_\chi \geq k^{-2^t}$.

Lemma 4.1 (Slices)**.** *Let $1 \leq j \leq m$ and $A \subseteq [j-1]$ with $|A| = t$. At least M^{m-t-1} vectors $\beta \in [M]^m$ are good for (j, A).*

Proof. Write $A = \{a_{(1)} < \cdots < a_{(t)}\}$ and let $K = [m] \setminus (A \cup \{j\})$, a set of $m - t - 1$ indices. We show that every choice of $(\beta_i)_{i \in K} \in [M]^K$ extends to a vector β that is good for (j, A). Different choices give different vectors, which proves the lemma.

Fix $(\beta_i)_{i \in K}$, let $R = \sum_{i<j,\, i \notin A} \beta_i$, which involves only indices in K, and put $r = R + 1$. Since $R \leq (j-1-t)M \leq (m-2)M$, we have $r + M \leq (m-1)M + 1 \leq n$. Proposition 3.2 gives u and $a_1, \dots, a_t$ with $\rho_\chi(u; a) \geq k^{-2^t}$, where $z = u - r$ satisfies $0 \leq z \leq M - t \leq M - 1$ and $1 \leq a_i \leq M$. Set

$$\beta_j = z + 1, \qquad \beta_{a_{(i)}} = a_i \quad (1 \leq i \leq t).$$

These values lie in $[M]$, and the shape (4.1) becomes $(R + z + 1; a_1, \dots, a_t) = (u; a_1, \dots, a_t)$. So β is good for (j, A). □

Lemma 4.2 (Averaging)**.** *There are $\sigma \in S_n$ and $\beta \in [M]^m$ such that at least*

$$\frac{|\mathcal{P}|}{k^{2^t} M^{t+1}}$$

of the cubes $Q_{\sigma,\beta}(j, A)$, $(j, A) \in \mathcal{P}$, are monochromatic.

Proof. Count the triples $(\sigma, \beta, (j, A)) \in S_n \times [M]^m \times \mathcal{P}$ for which $Q_{\sigma,\beta}(j, A)$ is monochromatic. For fixed $(j, A) \in \mathcal{P}$ and β good for (j, A), the number of suitable σ is $n!$ times the value of ρ_χ at the shape (4.1), which is at least $n!\, k^{-2^t}$. By Lemma 4.1 the number of triples is at least $|\mathcal{P}|\, M^{m-t-1}\, n!\, k^{-2^t}$. There are $n!\, M^m$ pairs (σ, β), so one of them occurs in at least $|\mathcal{P}|\, k^{-2^t} M^{-t-1}$ triples. □

Proof of Theorem 1.1. Let $c_{k,t}$ be as in the statement. Since $M \geq t+1$ we have $(4M^2)^{t+1} \geq (4M(t+1))^{t+1}$, so for $1 \leq n < 4M(t+1)$ the bound $F_k(n) \geq 1 > c_{k,t} n^{t+1}$ is trivial. Let $n \geq 4M(t+1)$.

Take σ, β as in Lemma 4.2 and apply Proposition 2.5 to the blocks $\sigma(Y_1^\beta), \ldots, \sigma(Y_m^\beta)$, which are pairwise disjoint and nonempty, and to the set $\mathcal{S} \subseteq \mathcal{P}$ of pairs whose cube is monochromatic. Every k-colouring therefore has a monochromatic union-closed family with at least

$$\frac{|\mathcal{P}|}{k^{2^{t+1}} M^{t+1}} \tag{4.2}$$

members. It remains to bound $|\mathcal{P}| = (m-s)\binom{s}{t}$ from below. Since $n - 1 < M(m+1)$ and $m \leq 2s+1$, we have $n \leq 2M(s+1)$. Together with $n \geq 4M(t+1)$ this gives $s \geq 2t+1$, so $s \geq 1$ and therefore $n \leq 4Ms$. Also $m - s \geq s$, and $\binom{s}{t} \geq (s/t)^t$ because $s \geq t$. Hence

$$|\mathcal{P}| \;\geq\; \frac{s^{t+1}}{t^t} \;\geq\; \frac{n^{t+1}}{t^t (4M)^{t+1}},$$

and (4.2) is at least $c_{k,t}\, n^{t+1}$.

For the last assertion let $k = 2$, so that $\log_2 M = 2\log_2 t + 2^t$ and

$$\begin{aligned} \log_2 \frac{1}{c_{2,t}} &= 2^t + 1 + t\log_2 t + (t+1)(2 + 2\log_2 M) \\ &= (2t+3)\, 2^t + (5t+4)\log_2 t + 2t + 3. \end{aligned}$$

Since $\log_2 t \leq t-1$ and $2^t \geq 2t$, the last two terms are at most $5t^2 + t - 1 \leq 2t(t+1)^2 \leq (t+1)^2 2^t$, so $\log_2(1/c_{2,t}) \leq (t+2)^2 2^t$. If $\log_2 n \geq (t+2)^2 2^t$, then $c_{2,t} n \geq 1$ and $F(n) \geq c_{2,t} n^{t+1} \geq n^t$. □

5. The upper bound and the random colouring

A family $A_1, \ldots, A_d$ of subsets of $[n]$ is an *independent system* if each A_i has a *private element*, that is, some $x_i \in A_i$ lying in no $A_{i'}$ with $i' \neq i$. A set $X \subseteq [n]$ is *shattered* by a family $\mathcal{F}$ if $\{A \cap X : A \in \mathcal{F}\} = 2^X$, and the *Vapnik–Chervonenkis dimension* $\mathrm{vc}(\mathcal{F})$ is the largest size of a shattered set.

Lemma 5.1. *Let $d \geq 0$ and $\mathcal{F} \subseteq 2^{[n]}$.*

(a) *If $A_1, \ldots, A_d$ is an independent system, the $2^d - 1$ unions $\bigcup_{i \in I} A_i$, $\emptyset \neq I \subseteq [d]$, are pairwise distinct.*

(b) *If $\mathcal{F}$ shatters a set of size d, then $\mathcal{F}$ contains an independent system of d sets.*
(c) (Sauer [Sa72], Shelah [Sh72]) *If* $\mathrm{vc}(\mathcal{F}) \le d$, *then* $|\mathcal{F}| \le \sum_{j=0}^{d} \binom{n}{j}$.

Proof. (a) The private element x_i lies in $\bigcup_{i' \in I} A_{i'}$ exactly when $i \in I$, so the union determines I. (b) If $X = \{x_1, \dots, x_d\}$ is shattered, choose $A_i \in \mathcal{F}$ with $A_i \cap X = \{x_i\}$; then x_i is a private element of A_i. (c) This is the Sauer–Shelah lemma. □

Proof of Theorem 1.2. Let χ be a uniformly random 2-colouring of $2^{[n]}$, the colours of the 2^n sets being independent and uniform. Call an ordered d-tuple $(A_1, \dots, A_d)$ of subsets of $[n]$ *bad* if it is an independent system and the $2^d - 1$ sets $\bigcup_{i \in I} A_i$, $\emptyset \ne I \subseteq [d]$, all have the same colour. By Lemma 5.1(a) these sets are distinct, so a fixed independent tuple is bad with probability $2^{1-(2^d-1)} = 2^{2-2^d}$. There are at most 2nd tuples, and $2^{nd} \cdot 2^{2-2^d} < 1$ by hypothesis, so some colouring has no bad tuple. Fix it.

Let $\mathcal{F}$ be a monochromatic union-closed family. If $\mathcal{F}$ contained an independent system of d sets, all unions of its members would lie in $\mathcal{F}$ and have the same colour, and the system would be bad. So $\mathcal{F}$ contains no independent system of d sets, $\mathrm{vc}(\mathcal{F}) \le d-1$ by Lemma 5.1(b), and $|\mathcal{F}| \le \sum_{j<d} \binom{n}{j}$ by Lemma 5.1(c).

For the second assertion let $n \ge 2$, and put $a = \lfloor \log_2 n \rfloor$ and $b = \lfloor \log_2 a \rfloor$, so that $a \ge 1$, $n < 2^{a+1}$, $a < 2^{b+1}$ and $b + 1 \le a$. For $d = a + b + 3$ we get

$$2^d = 2^{a+1} \cdot 2 \cdot 2^{b+1} \;\ge\; 2(n+1)(a+1),$$

whereas $nd + 2 = n(a+1) + n(b+2) + 2 \le 2n(a+1) + 2 < 2(n+1)(a+1)$, because $b + 2 \le a + 1$ and $a \ge 1$. So $2^d > nd + 2$, and the first assertion gives $F(n) \le \sum_{j<d} \binom{n}{j} \le \sum_{j<d} n^j \le d\, n^{d-1}$. Finally $a \le \log_2 n$ and $b \le \log_2 a \le \log_2 \log_2 n$, so $d \le \log_2 n + \log_2 \log_2 n + 3$. □

Figure 7 shows the size of the bound of Theorem 1.2 as a function of n and d. Taking for d the least admissible value $d(n)$ gives a bound valid for every n, and $d(n) = \log_2 n + \log_2 \log_2 n + O(1)$ as $n \to \infty$.

Proof of Corollary 1.3. (i) Let $n \ge 2$, so that $\ell = \log_2 \log_2 n \ge 0$. If $\omega(n) \le 0$ there is nothing to prove, since $F(n) \ge 1$. Otherwise $x = \ell - 2\log_2(\ell+2) > 1$, and the integer $t = \lfloor x \rfloor$ satisfies $t \ge 1$ and $\omega(n) = x - 1 < t \le x$. As $t \le \ell$ we have $\log_2(t+2) \le \log_2(\ell+2)$, and so

$$\log_2\big((t+2)^2 2^t\big) = t + 2\log_2(t+2) \;\le\; x + 2\log_2(\ell+2) = \ell,$$

that is, $(t+2)^2 2^t \le \log_2 n$. Theorem 1.1 gives $F(n) \ge n^t \ge n^{\omega(n)}$. As $\ell \to \infty$ and $\log_2(\ell+2) = o(\ell)$, we have $\omega(n) \to \infty$.

(ii) Put $\varepsilon(n) = F(n)^{1/n} - 1 + 1/n$ for $n \ge 1$. Then $(1+\varepsilon(n))^n = (F(n)^{1/n} + 1/n)^n > F(n)$. By Theorem 1.2, $\log F(n) = O((\log n)^2)$, so $F(n)^{1/n} \to 1$ and $\varepsilon(n) \to 0$.

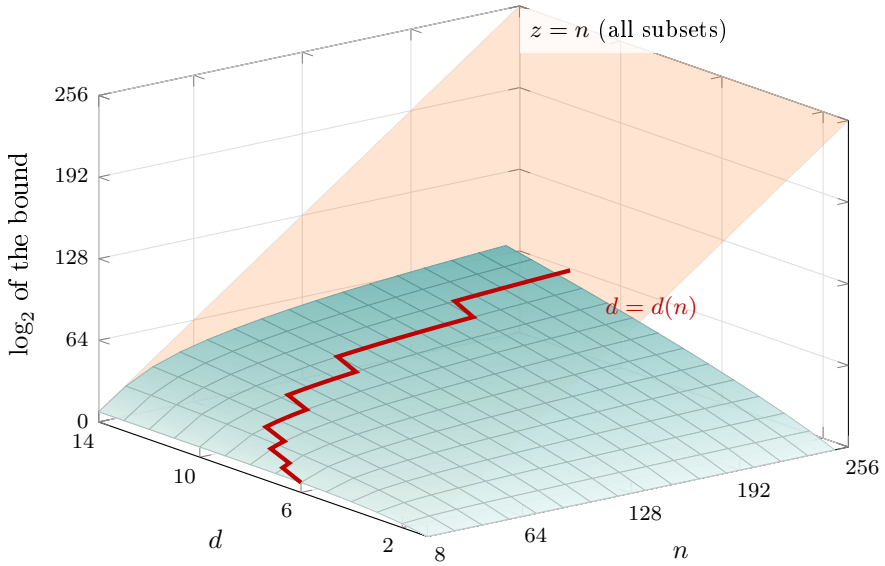


FIGURE 7. The surface $\log_2 \sum_{j<d} \binom{n}{j}$ over $8 \le n \le 256$ and $1 \le d \le 14$, with the plane $z = n$ of the trivial bound 2^n. The red curve follows the least admissible $d = d(n)$ of Theorem 1.2, along which the bound stays far below the plane.

(iii) Let $n \ge 2$ and write $\lambda = \log_2 n$ and $D = \lambda + \log_2 \lambda + 3$. Theorem 1.2 gives $n^{\omega(n)} \le F(n) \le D\, n^{D-1}$, hence

$$\omega(n) \;\le\; D - 1 + \frac{\log_2 D}{\log_2 n} \;=\; \lambda + O(\log \lambda),$$

which is at most $(1+\varepsilon)\lambda$ for all large n.

(iv) For every $t \ge 1$, Theorem 1.1 gives $F_k(n) \ge c_{k,t} n^{t+1} \ge n^t$ as soon as $n \ge 1/c_{k,t}$. Let $\omega_k(n)$ be the largest integer $t \le n$ with $n^t \le F_k(n)$; the value $t = 0$ qualifies because $F_k(n) \ge 1$. Then $\omega_k(n) \ge t$ whenever $n \ge \max(t, 1/c_{k,t})$, so $\omega_k(n) \to \infty$. For the second statement take $\varepsilon_k = \varepsilon$ from (ii) and use $F_k(n) \le F(n)$. □

The random colouring does contain large monochromatic union-closed families. How large they are indicates how much room is left between Theorems 1.1 and 1.2.

Proposition 5.2. *Let $n \ge 1$. The average, over all 2-colourings χ of $2^{[n]}$, of the largest size of a monochromatic union-closed family is at least*

$$\sum_{j=0}^{n} \binom{n}{j} 2^{1-2^j}.$$

If $1 \le j \le n$ and $2^{2^j} \le n$, this average is at least $(2/n)(n/j)^j$.

Proof. Let $c = \chi([n])$ and let $\mathcal{Y}$ be the set of $Y \subseteq [n]$ such that $\chi([n] \setminus Y') = c$ for every $Y' \subseteq Y$. Then $\mathcal{Y}$ is closed under taking subsets and under intersection, so $\mathcal{F} = \{[n] \setminus Y : Y \in \mathcal{Y}\}$ is monochromatic and union-closed, because $([n]\setminus Y)\cup([n]\setminus Y') = [n]\setminus(Y\cap Y')$. A set Y of size j lies in $\mathcal{Y}$ exactly when the $2^j - 1$ distinct sets $[n]\setminus Y'$ with $\emptyset \neq Y' \subseteq Y$ all have colour c, which happens for a proportion 2^{1-2^j} of all colourings. Summing over Y gives the average of $|\mathcal{F}|$, which is the first bound. For the second, the term of index j is at least $\binom{n}{j}2^{1-2^j} \geq (n/j)^j \cdot 2/n$, since $\binom{n}{j} \geq (n/j)^j$ and $2^{2^j} \leq n$. □

For $n \geq 4$ and $j = \lfloor \log_2 \log_2 n \rfloor$ the conditions of Proposition 5.2 hold, and $(2/n)(n/j)^j \geq n^{\log_2 \log_2 n - 2 - o(1)}$ because $j > \log_2 \log_2 n - 1$ and $j^j = n^{o(1)}$. So for the random colouring the exponent ω is at least about $\log_2 \log_2 n$, while Theorem 1.2, which is also proved with the random colouring, allows it to be as large as $\log_2 n$. The next two statements turn the upper bound for the random colouring into a counting problem.

Lemma 5.3. *If $\mathcal{F} \subseteq 2^{[n]}$ is union-closed and $N \leq |\mathcal{F}|$, then $\mathcal{F}$ contains a union-closed subfamily with exactly N members.*

Proof. It suffices to remove one member when $|\mathcal{F}| > N$. Let A be a member of least size. If $B, C \in \mathcal{F} \setminus \{A\}$ satisfied $B \cup C = A$, then $B \subseteq A$ and $|B| \geq |A|$ would give $B = A$. So $\mathcal{F} \setminus \{A\}$ is union-closed. □

Proposition 5.4. *Let $U(n, N)$ be the number of union-closed families $\mathcal{F} \subseteq 2^{[n]}$ with $|\mathcal{F}| = N$. If $U(n, N) < 2^{N-1}$, then $F(n) < N$.*

Proof. A fixed family of N distinct sets is monochromatic under exactly $2 \cdot 2^{2^n - N}$ of the 2^{2^n} colourings. The colourings under which some union-closed family of N sets is monochromatic are therefore at most $U(n, N)\, 2^{2^n + 1 - N} < 2^{2^n}$ in number, and some colouring has no monochromatic union-closed family of exactly N members. By Lemma 5.3 it has none with N or more members, since a subfamily of a monochromatic family is monochromatic. Hence $F(n) \leq N - 1$. □

Proposition 5.4 is the first-moment method for a uniformly random colouring, and Corollary 1.3(i) shows where it must fail: if $N \leq n^{\omega(n)} \leq F(n)$, then $U(n, N) \geq 2^{N-1}$. It fails much further up. For $m, k \geq 0$ write $S_k(m) = \sum_{j<k} \binom{m}{j}$.

Proposition 5.5.

(i) *If $m \leq n$ and $k, g \geq 0$, then $U\big(n, S_k(m) + g\big) \geq \binom{\binom{m}{k}}{g}$.*

(ii) *If $n \geq 206$, $N \geq 1$ and $5N \leq \binom{n}{\lfloor n/12 \rfloor}$, then $U(n, N) \geq 2^{N-1}$. In particular $U(n, N) \geq 2^{N-1}$ whenever $n \geq 206$ and $1 \leq N \leq 2^{n/4}$.*

Proof. (i) Fix $W \subseteq [n]$ with $|W| = m$, let $\mathcal{L}$ be the family of subsets of W with fewer than k elements, so that $|\mathcal{L}| = S_k(m)$, and let $\mathcal{G}$ be a family of

g subsets of W with exactly k elements. Put $\mathcal{F}_\mathcal{G} = \{W \setminus X : X \in \mathcal{L} \cup \mathcal{G}\}$. The map $X \mapsto W \setminus X$ is injective on subsets of W and $\mathcal{L} \cap \mathcal{G} = \emptyset$, so $|\mathcal{F}_\mathcal{G}| = S_k(m)+g$. The family $\mathcal{F}_\mathcal{G}$ is union-closed, because $(W\setminus X)\cup(W\setminus Y) = W \setminus (X\cap Y)$ and $X\cap Y \in \mathcal{L}\cup\mathcal{G}$ for $X, Y \in \mathcal{L}\cup\mathcal{G}$: this is clear if $X = Y$, and otherwise $X \cap Y$ has fewer than k elements, since X and Y have at most k elements and, if both have exactly k, then $X \cap Y$ is a proper subset of X. The members of $\mathcal{F}_\mathcal{G}$ with $m - k$ elements are the sets $W \setminus X$, $X \in \mathcal{G}$, so $\mathcal{G}$ is determined by $\mathcal{F}_\mathcal{G}$, and the $\binom{\binom{m}{k}}{g}$ choices of $\mathcal{G}$ give distinct families.

(ii) We use two inequalities. First,

$$a^g \le \binom{M}{g} \qquad \text{whenever } (a+1)g \le M+1, \tag{5.1}$$

by induction on g, since $\binom{M}{g+1}(g+1) = \binom{M}{g}(M-g)$ and $a(g+1) \le M-g$. Second, if $9k \le m$ and $j < k$, then $(m-j)/(j+1) \ge 8$, so $8\binom{m}{j} \le \binom{m}{j+1}$, and by induction on j

$$7S_j(m) \le \binom{m}{j} \qquad (j \le k,\ 9k \le m). \tag{5.2}$$

One window. Let $m \le n$ and $S = S_k(m)$, and let N satisfy $S \le N$, $2S \le N+1$ and $5(N-S) \le \binom{m}{k}+1$. With $g = N - S$, part (i) and (5.1) for $a = 4$ give $U(n,N) \ge 4^g \ge 2^{N-1}$, because $N \le 2g+1$.

One level. Let $k \ge 1$ and $9k \le m \le n$, and let N satisfy $2S_k(9k) \le N+1$ and $5(N - S_k(m)) \le \binom{m}{k}+1$. We show $U(n,N) \ge 2^{N-1}$ by induction on m. For $m = 9k$ the window applies, since $S_k(9k) \ge 1$. For $m+1$, the window at $m+1$ applies if $2S_k(m+1) \le N+1$. Otherwise $N \le 2S_k(m+1) - 2 = 2S_k(m) + 2S_{k-1}(m) - 2$ by Pascal's rule. By (5.2), $7S_k(m) \le \binom{m}{k}$ and $56S_{k-1}(m) \le 8\binom{m}{k-1} \le \binom{m}{k}$, so $5(N-S_k(m)) \le 5S_k(m)+10S_{k-1}(m) \le \binom{m}{k}$, and the case m applies.

From one level to the next. Let $k \ge 1$, $n \ge 206$ and $12(k+1) \le n$, and put $a = 9(k+1)$. Then $S_{k+1}(a) = S_k(a) + \binom{a}{k} \le \frac{8}{7}\binom{a}{k}$ by (5.2). Comparing the factors $(a-i)n \le (n-i)a$ gives $\binom{a}{k}n^k \le \binom{n}{k}a^k$. Moreover $80a^k \le 7n^k$: for $k \ge 9$ because $n \ge \frac{4}{3}a$ and $7(4/3)^k \ge 80$, and for $k \le 8$ because $80\,(9k+9)^k \le 7 \cdot 206^k$. Hence $80\binom{a}{k} \le 7\binom{n}{k}$ and

$$10\,S_{k+1}\big(9(k+1)\big) \le \binom{n}{k}. \tag{5.3}$$

All levels. Let $n \ge 206$ and $K = \lfloor n/12 \rfloor$. We show by induction on $k = 1, \dots, K$ that every $N \ge 1$ with $5(N - S_k(n)) \le \binom{n}{k}+1$ satisfies $U(n,N) \ge 2^{N-1}$. For $k = 1$ the window at $m = n$ applies, as $S_1(n) = 1$. For $k+1 \le K$, if $2S_{k+1}(9(k+1)) \le N+1$ the level $k+1$ with $m = n$ applies; otherwise $5N < 10S_{k+1}(9(k+1)) \le \binom{n}{k}$ by (5.3), and the case k applies. For $k = K$ every $N \ge 1$ with $5N \le \binom{n}{K}$ qualifies.

Finally let $1 \le N \le 2^{n/4}$. By (5.1) with $a = 11$, $\binom{n}{K} \ge 11^K$, and $K \ge 17$ gives $625 \cdot 2^n \le 625 \cdot 2^{12K+11} \le 11^{4K}$. Hence $(5N)^4 \le 625 \cdot 2^n \le \binom{n}{K}^4$, that is, $5N \le \binom{n}{K}$. □

Corollary 5.6. *For $n \ge 206$, Proposition 5.4 gives $F(n) < N$ only for $N > 2^{n/4}$. The answer to Question 9.2 is negative for every constant C: for all large n there is $N \ge n^{C \log_2 \log_2 n}$ with $U(n, N) \ge 2^{N-1}$.*

Proof. The first statement is Proposition 5.5(ii). For the second take $N = 2^{\lfloor n/4 \rfloor}$; since $(\log n)^2 = o(n)$, we have $n^{C \log_2 \log_2 n} \le N$ for all large n. □

The negative answer to Question 9.2 was first obtained in the trureturing repository [Tr26], for $n = 6r$, from the families made of all sets with more than $5r$ elements and one set from each of $\lfloor \binom{6r}{r}/4 \rfloor$ disjoint quadruples of $5r$-sets. Corollary 5.6 gives the same answer for every $n \ge 206$ and every $N \le 2^{n/4}$.

The families in the proof of Proposition 5.5 have all their members among the sets with at least $m-k$ elements, and their number exceeds 2^{N-1} because $\mathcal{G}$ can be any family of k-sets. An upper bound for $F(n)$ therefore has to use the structure of union-closed families, as Theorem 1.2 does through the Vapnik–Chervonenkis dimension, and not only their number.

6. Cardinality colourings, Hilbert cubes and arithmetic progressions

In this section $k \ge 2$ and $\chi(S) = g(|S|)$ for a function $g : \{0, 1, \dots, n\} \to [k]$. A t-cube of shape $(u; a_1, \dots, a_t)$ consists of sets whose sizes are the elements of $H(u; a_1, \dots, a_t)$, and every permutation preserves sizes, so

$$\rho_\chi(u; a_1, \dots, a_t) = \begin{cases} 1 & \text{if } g \text{ is constant on } H(u; a_1, \dots, a_t), \\ 0 & \text{otherwise.} \end{cases} \tag{6.1}$$

Figure 8 shows a monochromatic Hilbert cube of dimension two.

Since ρ_χ takes only the values 0 and 1 here, Proposition 3.2 turns into Hilbert's lemma with an explicit window.

Corollary 6.1. *Let $k \ge 2$, $t \ge 1$, $M = t^2 k^{2^t}$ and $r \ge 0$. Every $g : \{r, r+1, \dots, r+M\} \to [k]$ is constant on some Hilbert cube $H(u; a_1, \dots, a_t)$ with $a_i \ge 1$, $u \ge r$ and $u + a_1 + \cdots + a_t \le r + M$.*

Proof. Extend g arbitrarily to $\{0, \dots, r-1\}$ and apply Proposition 3.2, with $n = r + M$, $L = L_k(t)$ and $M = M_k(t)$, to the cardinality colouring $S \mapsto g(|S|)$ of $2^{[r+M]}$. It gives a shape with $r \le u$, $(u - r) + \sum_i a_i \le M$ and $\rho_\chi(u; a) \ge k^{-2^t} > 0$, so $\rho_\chi(u; a) = 1$ and g is constant on $H(u; a)$ by (6.1). The elements of $H(u; a)$ lie between $u \ge r$ and $u + \sum_i a_i \le r + M$. □

$H(u; a_1, a_2) = \{u,\ u + a_1,\ u + a_2,\ u + a_1 + a_2\}$ with $u = 5$, $a_1 = 3$, $a_2 = 7$

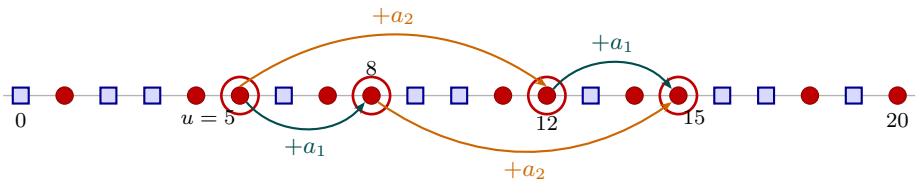


FIGURE 8. A colouring g of $\{0, 1, \ldots, 20\}$, red discs and blue squares, and a red Hilbert cube $H(5; 3, 7) = \{5, 8, 12, 15\}$. For the cardinality colouring $\chi(S) = g(|S|)$, every 2-cube of shape $(5; 3, 7)$ is red.

Proof of Theorem 1.5. Consider the cardinality colouring $\chi(S) = g(|S|)$ and the construction of Section 4, whose hypotheses hold since $n \geq 4M(t + 1)$ gives $s \geq n/(4M) \geq t + 1$. By (6.1), a vector β is good for (j, A) precisely when g is constant on the Hilbert cube formed by the sizes of the members of $Q_{\mathrm{id},\beta}(j, A)$, and these sizes are

$$|C_j \setminus Y_B| = \beta_1 + \cdots + \beta_j - \sum_{b \in B} \beta_b \qquad (B \subseteq A),$$

that is, the elements of $H_\beta(j, A)$. By Lemma 4.1, for every $(j, A) \in \mathcal{P}$ at least M^{m-t-1} vectors β make $H_\beta(j, A)$ monochromatic. Averaging over the M^m vectors, as in Lemma 4.2 but without the permutations, some β makes at least $|\mathcal{P}|/M^{t+1}$ of the cubes $H_\beta(j, A)$ monochromatic, and at least a proportion $1/k$ of these share a colour c. The estimate $|\mathcal{P}| \geq n^{t+1}/(t^t(4M)^{t+1})$ from the proof of Theorem 1.1 completes the proof. All elements of the cubes lie in $\{0, \ldots, n\}$, since $\beta_1 + \cdots + \beta_m \leq mM \leq n - 1$. □

Given β and c as in Theorem 1.5, the cubes $Q_{\mathrm{id},\beta}(j, A)$ attached to the pairs of colour c are monochromatic under χ, and Proposition 2.5 glues them into a single union-closed family of colour c. For cardinality colourings the proof of Theorem 1.1 is therefore entirely a statement about sums of a bounded sequence of integers.

There is also a direct route through arithmetic progressions, which gives Howorka's theorem.

Proposition 6.2. *Suppose that g is constant on the arithmetic progression $a, a + e, \ldots, a + (p - 1)e$, where $e \geq 1$, $p \geq 1$ and $a + (p - 1)e \leq n$. Put $q = \lfloor a/e \rfloor$ and $b = q + p - 1$. Then χ has a monochromatic union-closed family with $\sum_{i=0}^{p-1} \binom{b}{i}$ members and a monochromatic sublattice with 2^{p-1} members.*

Proof. Let $a_0 = a - qe$, so $0 \le a_0 < e$, and let $D = \{1, \dots, a_0\}$ and $X_i = \{a_0 + (i-1)e + 1, \dots, a_0 + ie\}$ for $1 \le i \le b$. These sets are pairwise disjoint subsets of $[n]$, because $a_0 + be = a + (p-1)e \le n$, and $|D| = a_0$ and $|X_i| = e$. The family

$$\mathcal{F} = \big\{ D \cup X_I : \ I \subseteq [b], \ |I| \ge q \big\}$$

is union-closed, since $D \cup X_I \cup D \cup X_{I'} = D \cup X_{I \cup I'}$ and $|I \cup I'| \ge q$ whenever $|I| \ge q$. A member with index set I has size $a_0 + |I|e = a + (|I| - q)e$, where $0 \le |I| - q \le p - 1$, so all members have the colour of the progression, and

$$|\mathcal{F}| = \sum_{i=q}^{b} \binom{b}{i} = \sum_{i=0}^{p-1} \binom{b}{i}$$

by the symmetry $\binom{b}{i} = \binom{b}{b-i}$. For the sublattice take $D' = \{1, \dots, a\}$ and $X'_i = \{a + (i-1)e + 1, \dots, a + ie\}$ for $1 \le i \le p - 1$. The $(p-1)$-cube $Q(D'; X'_1, \dots, X'_{p-1})$ is a sublattice with 2^{p-1} members, whose sizes $a + |I|e$ lie in the progression. □

For $k \ge 1$ and $p \ge 1$ let $W_k(p)$ be the least W such that every $g : \{0, 1, \dots, W-1\} \to [k]$ is constant on some progression $a, a+e, \dots, a+(p-1)e$ with $e \ge 1$. This number exists by van der Waerden's theorem.

Theorem 6.3 (van der Waerden [vdW27])**.** *For all $k \ge 1$ and $p \ge 1$ there is W such that every $g : \{0, 1, \dots, W-1\} \to [k]$ is constant on an arithmetic progression of length p with a positive common difference.*

The formal proof derives Theorem 6.3 from the Hales–Jewett theorem.[4]

Theorem 6.4 (Howorka)**.** *Let $p \ge 2$, let $W = W_k(p)$, and let $n \ge 2W$. Every cardinality k-colouring of $2^{[n]}$ has a monochromatic union-closed family with at least*

$$\binom{\lfloor n/(2W) \rfloor + p - 1}{p - 1} \ \ge \ \frac{\big(\lfloor n/(2W) \rfloor + 1\big)^{p-1}}{(p-1)!}$$

members.

Proof. The integers $n - W, n - W + 1, \dots, n - 1$ lie in $\{0, \dots, n\}$, and by the definition of W the function g is constant on a progression $a, a + e, \dots, a + (p-1)e$ among them, with $e \ge 1$. Here $a \ge n - W$ and $a + (p-1)e \le n - 1$, and

[4] Let N be such that every k-colouring of the words of length N over the alphabet $\{0, 1, \dots, p-1\}$ has a monochromatic combinatorial line, and put $W = N(p-1) + 1$. Colour a word by the value of g at the sum of its letters. A combinatorial line has a nonempty set E of active coordinates, which carry the letter i at its i-th point, and fixed letters elsewhere, with sum a say. The sums of its points are $a + i|E|$ for $0 \le i \le p-1$, all less than W, so a monochromatic line gives a monochromatic progression with common difference $|E|$.

$e \le (p-1)e \le W-1$ because $p \ge 2$. Since $n \ge 2W$ we have $n - W \ge \lfloor n/2 \rfloor$, and so

$$\left\lfloor \frac{a}{e} \right\rfloor \;\ge\; \left\lfloor \frac{n-W}{W} \right\rfloor \;\ge\; \left\lfloor \frac{\lfloor n/2 \rfloor}{W} \right\rfloor = \left\lfloor \frac{n}{2W} \right\rfloor.$$

Proposition 6.2 gives a monochromatic union-closed family with at least $\binom{b}{p-1}$ members, where $b = \lfloor a/e \rfloor + p - 1 \ge \lfloor n/(2W) \rfloor + p - 1$, and $\binom{b}{p-1}$ increases with b. Finally, with $N = \lfloor n/(2W) \rfloor$,

$$\binom{N+p-1}{p-1} = \frac{(N+1)(N+2)\cdots(N+p-1)}{(p-1)!} \;\ge\; \frac{(N+1)^{p-1}}{(p-1)!}. \qquad \square$$

For fixed p the bound of Theorem 6.4 is at least a constant multiple of n^{p-1}, and p is arbitrary. So for cardinality colourings the largest monochromatic union-closed family grows faster than every fixed power of n, which is Howorka's theorem as recorded in [Bl26]; for all colourings this is contained in Corollary 1.3. Proposition 6.2 also restricts the cardinality colourings that are good for f: if no monochromatic sublattice has more than m members, then g is constant on no progression of length p in $\{0, \dots, n\}$ with $2^{p-1} > m$.

7. The lattice function

This section proves Theorem 1.4. The lower bound improves the chain argument behind (1.1); the upper bounds come from a random colouring and two counts of sublattices.

7.1. **The lower bound.** We call the two colours red and blue. For a 2-colouring χ of 2^W, a *red–blue pair* is a pair $(\mathcal{R}, \mathcal{B})$ of sublattices of 2^W, either of which may be empty, such that every member of $\mathcal{R}$ is red and every member of $\mathcal{B}$ is blue. Its *size* is $|\mathcal{R}| + |\mathcal{B}|$. One of the two sublattices has at least half of these members, so a red–blue pair of size N yields a monochromatic sublattice with at least $\lceil N/2 \rceil$ members. The proof of (1.1) splits a maximal chain into its red and its blue members, which is a red–blue pair of size $n+1$. Three simple facts are used throughout.

(L1) A family of sets any two of which are comparable under inclusion is a sublattice.

(L2) If $\mathcal{L}_1$ and $\mathcal{L}_2$ are sublattices and $A \subseteq B$ for all $A \in \mathcal{L}_1$ and $B \in \mathcal{L}_2$, then $\mathcal{L}_1 \cup \mathcal{L}_2$ is a sublattice, since $A \cup B = B$ and $A \cap B = A$ for such A and B.

(L3) If $\mathcal{L}_1, \dots, \mathcal{L}_r$ are sublattices of $2^{W_1}, \dots, 2^{W_r}$ for pairwise disjoint sets $W_1, \dots, W_r$, then

$$\mathcal{L}_1 \otimes \cdots \otimes \mathcal{L}_r = \big\{ A_1 \cup \cdots \cup A_r : \; A_i \in \mathcal{L}_i \text{ for } i \in [r] \big\}$$

is a sublattice of $2^{W_1 \cup \cdots \cup W_r}$ with $|\mathcal{L}_1| \cdots |\mathcal{L}_r|$ members, because unions and intersections of such sets are formed separately inside each W_i.

The improvement over (1.1) is local: a stretch of twelve consecutive levels of a chain can always be traded for a red–blue pair with thirteen members.

Lemma 7.1 (Windows). *Every 2-colouring of $2^{[11]}$ has a red–blue pair of size at least 13.*

The proof distinguishes whether or not the colour of a set is determined by its size. Figure 9 illustrates the two cases.

Lemma 7.2 (Adjacent sets). *Let $m \geq 1$, and let χ be a 2-colouring of $2^{[m]}$ under which two sets of the same size have different colours. Then χ has a red–blue pair of size $m+2$.*

Proof. Among all pairs of sets of equal size and different colours choose S, T with $|S \setminus T|$ as small as possible. Suppose that $|S \setminus T| \geq 2$, pick $a \in S \setminus T$ and $b \in T \setminus S$, and put $U = (S \setminus \{a\}) \cup \{b\}$. Then $|U| = |S|$, $|S \setminus U| = 1$ and $|U \setminus T| = |S \setminus T| - 1$. If $\chi(U) \neq \chi(S)$, the pair S, U contradicts the choice of S, T; otherwise $\chi(U) \neq \chi(T)$, and the pair U, T does. Hence $|S \setminus T| = 1$, and after exchanging the names of S and T we may assume that S is red and T is blue.

Put $C = S \cap T$ and $D = S \cup T$, so that $|C| = |S| - 1$ and $|D| = |S| + 1$. Let $\mathcal{K}_1$ be a maximal chain of subsets of C, running from $\emptyset$ to C, and let $\mathcal{K}_2$ be a maximal chain of sets between D and $[m]$. They have $|C| + 1$ and $m - |D| + 1$ members, so $\mathcal{K} = \mathcal{K}_1 \cup \mathcal{K}_2$ has m members. Every member of $\mathcal{K}_1$ is contained in C and every member of $\mathcal{K}_2$ contains D. As $C \subset S \subset D$ and $C \subset T \subset D$, both $\mathcal{K} \cup \{S\}$ and $\mathcal{K} \cup \{T\}$ are chains, and neither S nor T belongs to $\mathcal{K}$, whose members have fewer than $|S|$ or more than $|S|$ elements. Let $\mathcal{R}$ consist of the red members of $\mathcal{K} \cup \{S\}$ and $\mathcal{B}$ of the blue members of $\mathcal{K} \cup \{T\}$. By (L1) both are sublattices, and $|\mathcal{R}| + |\mathcal{B}| = |\mathcal{K}| + 2 = m + 2$, since each member of $\mathcal{K}$ is counted once and S and T are counted once each. □

It remains to treat colourings with $\chi(S) = g(|S|)$, and we record g as the word $g(0)g(1)\cdots g(11)$ in the letters R (red) and B (blue). Let $u = u_0 u_1 \cdots u_\ell$ be such a word and let c be a letter. A sublattice $\mathcal{G}$ of 2^W, where $|W| = \ell$, *beats u in colour c* if $u_{|G|} = c$ for every $G \in \mathcal{G}$ and $\mathcal{G}$ has more members than there are indices i with $u_i = c$. The same sublattice beats, in the other colour, the word obtained from u by exchanging the two letters. The map $G \mapsto W \setminus G$ exchanges unions and intersections and sends a set of size i to one of size $\ell - i$, so it turns $\mathcal{G}$ into a sublattice that beats the reversed word $u_\ell \cdots u_1 u_0$ in colour c.

For positive integers $a_1, \ldots, a_r$ we write $\langle a_1, \ldots, a_r \rangle$ for a chain $\emptyset \subset Y_1 \subset Y_1 \cup Y_2 \subset \cdots \subset Y_1 \cup \cdots \cup Y_r$ built from pairwise disjoint sets with $|Y_i| = a_i$. In a product $\langle \cdots \rangle \otimes \cdots \otimes \langle \cdots \rangle$ the sets used by different factors are disjoint as well, as (L3) requires.

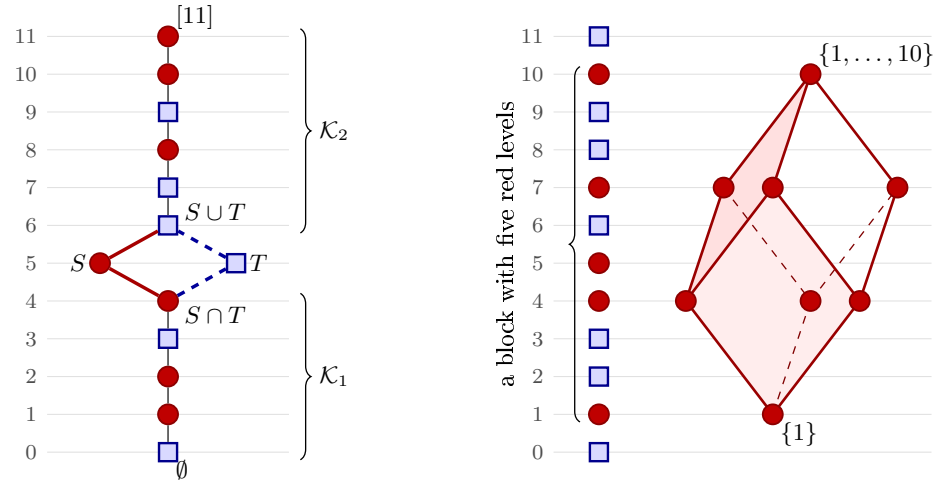


FIGURE 9. The two cases of Lemma 7.1, with red discs and blue squares and the levels of $2^{[11]}$ drawn horizontally. (a) Lemma 7.2: a maximal chain through $S \cap T$ and $S \cup T$ is shared by the two colours, S is added to the red sets and T to the blue sets, and the pair has 13 members. (b) The cardinality colouring given by the word RBBRRBRBBRB. Its block RBBRRBRBBR at the levels $1, \ldots, 10$ has five red letters, while the cube $\{\{1\} \cup X_I : I \subseteq [3]\}$ with $X_1 = \{2, 3, 4\}$, $X_2 = \{5, 6, 7\}$ and $X_3 = \{8, 9, 10\}$ has eight red members, at the levels 1, 4, 7 and 10.

Lemma 7.3 (Eight words)**.** *For each row of Table 1, the sublattice $\mathcal{G}$ listed there beats the word of that row in colour* R.

Proof. Each chain $\langle \cdots \rangle$ is a sublattice by (L1), so $\mathcal{G}$ is a sublattice of 2^W by (L3), and $|\mathcal{G}|$ is the product of the lengths of the chains. The sizes of the members of $\mathcal{G}$ are the sums of one size from each chain, which gives the fourth column. Every size listed there is a red index of the word, and $|\mathcal{G}|$ exceeds the number of red letters in each row. □

Lemma 7.4 (Words of length twelve)**.** *Every word of length* 12 *in the letters* R *and* B *contains, as a block of consecutive letters, one of the eight words of Table 1 or a word obtained from one of them by reversing it, by exchanging the two letters, or by both.*

Proof. Call the eight words and the words obtained from them in this way *forbidden*, and suppose that a word u of length 12 contains no forbidden block. A *run* of u is a maximal block of equal letters. Let the runs of u have lengths $r_1, \ldots, r_k$ from left to right, so that $r_1 + \cdots + r_k = 12$, and call

word	red indices	$\mathcal{G}$	sizes of members	$\lvert\mathcal{G}\rvert$	red letters
RRR	$0, 1, 2$	$\langle 1\rangle \otimes \langle 1\rangle$	$0, 1, 1, 2$	4	3
RBRBR	$0, 2, 4$	$\langle 2\rangle \otimes \langle 2\rangle$	$0, 2, 2, 4$	4	3
RBBRBBR	$0, 3, 6$	$\langle 3\rangle \otimes \langle 3\rangle$	$0, 3, 3, 6$	4	3
RRBBRRBBR	$0, 1, 4, 5, 8$	$\langle 1, 3\rangle \otimes \langle 4\rangle$	$0, 1, 4, 4, 5, 8$	6	5
RBRRBRBBR	$0, 2, 3, 5, 8$	$\langle 2, 3\rangle \otimes \langle 3\rangle$	$0, 2, 3, 5, 5, 8$	6	5
RRBRBBRRBR	$0, 1, 3, 6, 7, 9$	$\langle 3\rangle \otimes \langle 3\rangle \otimes \langle 3\rangle$	$0, 3, 3, 3, 6, 6, 6, 9$	8	6
RBRRBBRRBR	$0, 2, 3, 6, 7, 9$	$\langle 3\rangle \otimes \langle 3\rangle \otimes \langle 3\rangle$	$0, 3, 3, 3, 6, 6, 6, 9$	8	6
RBBRRBRBBR	$0, 3, 4, 6, 9$	$\langle 3\rangle \otimes \langle 3\rangle \otimes \langle 3\rangle$	$0, 3, 3, 3, 6, 6, 6, 9$	8	5

TABLE 1. Eight words and sublattices that beat them in colour R. In each row the ground set W is the union of the sets used by the chains, and its size is the length of the word minus one. The last two columns give the number of members of $\mathcal{G}$ and the number of red letters of the word.

the runs other than the first and the last *inner*. Since RRR and BBB are forbidden, every r_i is 1 or 2.

Several forbidden words consist of whole runs framed by a single letter on each side. For example, if three consecutive inner runs had length 1, then together with the last letter of the run before them and the first letter of the run after them they would form RBRBR or BRBRB. In the same way the forbidden words RBBRBBR, RBRRBRBBR, RBRRBBRRBR and RBBRRBRBBR, with their reversals, show that the sequence $v = (r_2, \dots, r_{k-1})$ of lengths of inner runs contains none of the blocks

(a) $1, 1, 1$ and $2, 1, 2$;
(b) $1, 2, 1, 1, 2$ and $2, 1, 1, 2, 1$;
(c) $1, 2, 2, 2, 1$;
(d) $2, 2, 1, 1, 2$ and $2, 1, 1, 2, 2$.

Two more facts concern the first and the last run.

(e) No four consecutive runs have length 2. Such runs contain eight letters, so another run precedes or follows them, and one letter of it completes RRBBRRBBR or a word obtained from it by reversing or exchanging letters.
(f) It is not the case that $r_1 = 2$, $(r_2, \dots, r_6) = (1, 1, 2, 2, 1)$ and $k \geq 7$. Otherwise the first nine letters of u and the first letter of the seventh run would form RRBRBBRRBR or BBRBRRBBRB.

As $r_1 + r_k \leq 4$, the entries of v add up to at least 8.

Suppose first that some entries 1 of v lie between two entries 2, and take a maximal block of such entries. By (a) the block is $1, 1$, so v contains $2, 1, 1, 2$. By (d) neither of these two entries 2 has a further entry 2 next to it in v,

so any entry of v just before or just after the block $2,1,1,2$ is a 1, and by (b) there is no such entry. Thus $v = (2,1,1,2)$, whose entries add up to 6, a contradiction.

Otherwise the entries 2 of v are consecutive, and v consists of a entries 1, then b entries 2, then c entries 1, where $a, c \le 2$ by (a) and $b \le 3$ by (e). From $a + 2b + c \ge 8$ we get $b \ge 2$. If $b = 3$ and $a, c \ge 1$, then v contains $1,2,2,2,1$, contrary to (c). If $b = 3$ and $a = 0$, then $c = 2$, the entries of v add up to exactly 8, so $r_1 = r_k = 2$ and the first four runs have length 2, contrary to (e); the case $c = 0$ is symmetric. If $b = 2$, then $a = c = 2$, again $r_1 = r_k = 2$, and $(r_1, \dots, r_8) = (2,1,1,2,2,1,1,2)$, contrary to (f). □

The length 12 in Lemma 7.4 cannot be lowered: the word RRBRBBRRBBR of length 11 contains no forbidden block.

Proof of Lemma 7.1. Let χ be a 2-colouring of $2^{[11]}$. If two sets of the same size have different colours, Lemma 7.2 with $m = 11$ gives a red–blue pair of size 13. Otherwise $\chi(S) = g(|S|)$ for a word $g = g(0)\cdots g(11)$. By Lemmas 7.4 and 7.3 and the two symmetries noted above, there are integers $0 \le a < b \le 11$, a colour c and a sublattice $\mathcal{G}$ of 2^W with $W = \{a+1, \dots, b\}$ that beats the block $g(a)g(a+1)\cdots g(b)$ in colour c. Write $S_i = [i]$ for $0 \le i \le 11$ and consider

$$\mathcal{L} = \{S_i : \ i < a, \ g(i) = c\} \ \cup \ \{S_a \cup G : \ G \in \mathcal{G}\} \ \cup \ \{S_i : \ i > b, \ g(i) = c\}.$$

The map $G \mapsto S_a \cup G$ is injective and preserves unions and intersections, so the middle family is a sublattice with $|\mathcal{G}|$ members. For $i < a$, $G \in \mathcal{G}$ and $i' > b$ we have $S_i \subseteq S_a \cup G \subseteq S_b \subseteq S_{i'}$, so $\mathcal{L}$ is a sublattice by (L1) and two applications of (L2), and its three parts are disjoint. Every member of $\mathcal{L}$ has colour c, because $|S_a \cup G| = a + |G|$. Since $\mathcal{G}$ has more members than there are indices i with $a \le i \le b$ and $g(i) = c$, the family $\mathcal{L}$ has more members than there are indices $0 \le i \le 11$ with $g(i) = c$. The sets S_i with $g(i) \ne c$ form a chain of the other colour, and together with $\mathcal{L}$ they give a red–blue pair of size at least 13. □

Stacking windows along a maximal chain gives the lower bound of Theorem 1.4.

Proposition 7.5. *Let $n \ge 1$ and $q = \lfloor (n+1)/12 \rfloor$. Every 2-colouring of $2^{[n]}$ has a red–blue pair of size at least $n + 1 + q$. Consequently $f(n) \ge \lceil (n + 1 + q)/2 \rceil$.*

Proof. Let χ be a 2-colouring of $2^{[n]}$ and write $S_i = [i]$. For $0 \le j < q$ let $W_j = \{12j + 1, \dots, 12j + 11\}$, a subset of $[n]$ because $12q \le n + 1$. The map $T \mapsto S_{12j} \cup T$ from 2^{W_j} to $2^{[n]}$ is injective and preserves unions and intersections. Applying Lemma 7.1 to the colouring $T \mapsto \chi(S_{12j} \cup T)$ of 2^{W_j} we obtain a red–blue pair $(\mathcal{R}_j, \mathcal{B}_j)$ of χ of size at least 13 whose members A

satisfy $S_{12j} \subseteq A \subseteq S_{12j+11}$, and hence $12j \le |A| \le 12j+11$. Let $\mathcal{C}$ be the chain of the sets S_i with $12q \le i \le n$, which has $n+1-12q$ members. For $j < j'$, every member of $\mathcal{R}_j \cup \mathcal{B}_j$ is contained in every member of $\mathcal{R}_{j'} \cup \mathcal{B}_{j'}$ and of $\mathcal{C}$, and is smaller than each of them. By (L1) and repeated use of (L2),

$$\mathcal{R} = \mathcal{R}_0 \cup \cdots \cup \mathcal{R}_{q-1} \cup \{A \in \mathcal{C} :\ A \text{ is red}\},$$
$$\mathcal{B} = \mathcal{B}_0 \cup \cdots \cup \mathcal{B}_{q-1} \cup \{A \in \mathcal{C} :\ A \text{ is blue}\}$$

form a red–blue pair. The unions are disjoint, so $|\mathcal{R}| + |\mathcal{B}| \ge 13q + (n+1-12q) = n+1+q$. □

For odd $n \ge 11$ and for $n \ge 23$ we have $\lceil (n+1+q)/2 \rceil > \lceil (n+1)/2 \rceil$, and since $q \ge (n-10)/12$ we have $(n+1+q)/2 \ge (13n+2)/24$. These are the lower bounds stated in Theorem 1.4.

7.2. **Counting sublattices.** The upper bounds rest on two counts of sublattices. The first uses Birkhoff's description of a sublattice through its least member and the least members containing the points; the second removes the points one at a time.

Let $\mathcal{L} \subseteq 2^{[n]}$ be a nonempty sublattice, and put $B = \bigcap_{A \in \mathcal{L}} A$ and $T = \bigcup_{A \in \mathcal{L}} A$, the least and the greatest member of $\mathcal{L}$. For $x \in T \setminus B$ let

$$A_x = \bigcap \{A \in \mathcal{L} :\ x \in A\},$$

the least member of $\mathcal{L}$ containing x, and for $X \subseteq T \setminus B$ put $A_X = \bigcup_{x \in X} A_x$.

Lemma 7.6. *Let $\mathcal{L}$ be a nonempty sublattice of $2^{[n]}$. Every $A \in \mathcal{L}$ equals $B \cup A_{A \setminus B}$, and $B \cup A_X \in \mathcal{L}$ for every $X \subseteq T \setminus B$. Hence $\mathcal{L} = \{B \cup A_X : X \subseteq T \setminus B\}$.*

Proof. Let $A \in \mathcal{L}$. Then $B \subseteq A \subseteq T$, and $A_x \subseteq A$ for every $x \in A \setminus B$, so $B \cup A_{A \setminus B} \subseteq A$; conversely every $x \in A$ lies in B or in A_x. The set $B \cup A_X$ is a union of members of $\mathcal{L}$, or equals B when $X = \emptyset$. □

Lemma 7.6 is the form that Birkhoff's representation theorem for finite distributive lattices [Bi37] takes for sublattices of $2^{[n]}$.[5] It shows that $\mathcal{L}$ is determined by B and the map $x \mapsto A_x$. This description uses about n^2 bits for every sublattice, however small. The second count is sensitive to the size of the sublattice and comes from deleting the largest point.

For $1 \le k \le n$ and a family $\mathcal{L} \subseteq 2^{[k]}$ put

$$\mathcal{L}^- = \big\{A \setminus \{k\} :\ A \in \mathcal{L}\big\} \subseteq 2^{[k-1]}.$$

[5]The sets A_x are exactly the members of $\mathcal{L}$ other than B that are not the union of two smaller members: A_x is not such a union because one of the two parts would contain x and hence all of A_x, and by Lemma 7.6 any other member is the union of the sets A_x it contains, all of them smaller. Birkhoff's theorem identifies a finite distributive lattice with the down-sets of the poset of these elements.

If some member of a sublattice $\mathcal{L} \subseteq 2^{[k]}$ contains k, let A_k be the least such member, and if some member avoids k, let Z_k be the union of all members avoiding k, which is the greatest of them.

Lemma 7.7 (Removing a point)**.** *Let $1 \le k \le n$ and let $\mathcal{L} \subseteq 2^{[k]}$ be a nonempty sublattice. Then $\mathcal{L}^-$ is a nonempty sublattice of $2^{[k-1]}$ with $|\mathcal{L}^-| \le |\mathcal{L}|$, the sets $A_k \setminus \{k\}$ and Z_k belong to $\mathcal{L}^-$ whenever they are defined, and*

$$\mathcal{L} = \big\{C \cup \{k\} : \ C \in \mathcal{L}^-, \ A_k \setminus \{k\} \subseteq C\big\} \ \cup \ \big\{C \in \mathcal{L}^- : \ C \subseteq Z_k\big\},$$

where the first family is empty if no member of $\mathcal{L}$ contains k and the second is empty if every member contains k.

Proof. Removing k commutes with union and intersection, so $\mathcal{L}^-$ is a sublattice, and it has at most $|\mathcal{L}|$ members. The set A_k is an intersection and Z_k a union of members of $\mathcal{L}$, so both lie in $\mathcal{L}$; hence $A_k \setminus \{k\} \in \mathcal{L}^-$, and $Z_k = Z_k \setminus \{k\} \in \mathcal{L}^-$.

Let $A \in \mathcal{L}$. If $k \in A$, then $A_k \subseteq A$, so $C = A \setminus \{k\}$ contains $A_k \setminus \{k\}$ and $A = C \cup \{k\}$. If $k \notin A$, then $A \subseteq Z_k$ and $A = A \setminus \{k\} \in \mathcal{L}^-$. Conversely, let $C = A \setminus \{k\}$ with $A \in \mathcal{L}$. If $A_k \setminus \{k\} \subseteq C$, then $A \cup A_k$ is a member of $\mathcal{L}$, and it equals $C \cup \{k\}$. If $C \subseteq Z_k$, then $A \cap Z_k$ is a member of $\mathcal{L}$, and it equals C because $k \notin Z_k$. □

Lemma 7.8. (a) *The number of nonempty sublattices of $2^{[n]}$ is at most $2^{n(n+1)}$.*

(b) *For $m \ge 1$, the number of nonempty sublattices of $2^{[n]}$ with at most m members is at most $(m+1)^{2n}$.*

Proof. (a) By Lemma 7.6, $\mathcal{L}$ is determined by B and the map $x \mapsto A_x$, extended by $A_x = B$ for $x \notin T \setminus B$. There are 2^n choices for B and $(2^n)^n$ for the map.

(b) For $0 \le k \le n$ let N_k be the number of nonempty sublattices of $2^{[k]}$ with at most m members. The only nonempty family of subsets of the empty set is $\{\emptyset\}$, so $N_0 = 1$. Let $k \ge 1$. By Lemma 7.7, the map $\mathcal{L} \mapsto \mathcal{L}^-$ sends the sublattices counted by N_k to sublattices counted by N_{k-1}, and $\mathcal{L}$ is determined by $\mathcal{L}^-$ together with the two sets $A_k \setminus \{k\}$ and Z_k. Each of these is a member of $\mathcal{L}^-$ or is undefined, which leaves at most $|\mathcal{L}^-| + 1 \le m + 1$ possibilities for each. Hence $N_k \le (m+1)^2 N_{k-1}$, and $N_n \le (m+1)^{2n}$. □

Proof of Theorem 1.4. The lower bounds follow from Proposition 7.5, as shown after its proof. For the upper bounds we use that the number of 2-colourings under which a fixed family of N distinct sets is monochromatic is $2 \cdot 2^{2^n - N}$.

For the first bound put $m_1 = n^2 + n + 2$. By Lemma 7.8(a), the colourings under which some sublattice with at least m_1 members is monochromatic number at most $2^{n(n+1)} \cdot 2 \cdot 2^{2^n - m_1} = 2^{2^n - 1}$. So some colouring has no such sublattice, and $f(n) \le m_1 - 1 = n^2 + n + 1$.

For the second bound recall $L = \lfloor \log_2 n \rfloor + 1$ and $K = \lfloor \log_2 L \rfloor + 1$, so that $n + 1 \le 2^L$, $L + 1 \le 2^K$ and $K \le L$. For $w \ge 0$ put

$$E(w) = 2n(w+1) + w + 3.$$

The function E is increasing, and $E(w) \le 2^w$ for $w \ge L + K + 3$. Indeed,

$$E(L+K+3) = 2n(L+K+4) + L + K + 6 \le 4nL + 8n + 2L + 6$$
$$\le 8(n+1)(L+1) \le 2^{L+K+3},$$

and $E(w+1) = E(w) + 2n + 1 \le 2E(w)$. Put $m_2 = E(L+K+2) = 2n(L+K+3) + L + K + 5$. Every sublattice $\mathcal{L}$ with $|\mathcal{L}| \ge m_2$ and $\lfloor \log_2 |\mathcal{L}| \rfloor = w$ satisfies $|\mathcal{L}| \ge E(w)$: for $w \le L + K + 2$ because $E(w) \le m_2$, and for $w \ge L + K + 3$ because $E(w) \le 2^w \le |\mathcal{L}|$. Such a sublattice has at most $2^{w+1} - 1$ members, so by Lemma 7.8(b) there are at most $(2^{w+1})^{2n} = 2^{2n(w+1)}$ of them for each w, and each is monochromatic under at most $2^{2^n + 1 - E(w)}$ colourings. The colourings under which some sublattice with at least m_2 members is monochromatic therefore number at most

$$\sum_{w=0}^{n} 2^{2n(w+1)}\, 2^{2^n + 1 - E(w)} = 2^{2^n} \sum_{w=0}^{n} 2^{-(w+2)} \;<\; 2^{2^n - 1},$$

the sum running up to $w = n$ because a sublattice has at most 2^n members. So some colouring has no such sublattice, and $f(n) \le m_2 - 1 = 2n(L+K+3) + L + K + 4$.

As $L \le \log_2 n + 1$ and $K \le \log_2(\log_2 n + 1) + 1$, the second bound is $2n \log_2 n + O(n \log \log n)$. The last assertion of the theorem follows from Theorem 1.1 with $t > C$. □

The first upper bound is the smaller one for $n \le 21$ and the second for $n \ge 22$. Both come from the random colouring, and the remaining gap, a factor of order $\log n$ between them and the lower bound of Proposition 7.5, is the subject of Question 9.3.

8. Formal verification

The results of Sections 2–7, except Lemmas 7.3 and 7.4, have been formalised in Lean 4 [MU21], version 4.34.1, with the Mathlib library [Mat20] at the release of the same version number. The development contains no unproved steps, and for each statement listed below Lean reports that the proof uses only the three standard axioms of its logic, namely propositional extensionality, the axiom of choice and the soundness of quotient types.[6]

In the formalisation the ground set is the type of n elements, a k-colouring is a function from its subsets to a finite type with k elements, and $F_k(n)$, $F(n)$ and $f(n)$ are defined literally as the minimum over all colourings of the largest

[6] These axioms are part of the foundations on which Mathlib is built. In particular, no statement relies on an additional axiom or on compiled code run in place of a proof.

size of a monochromatic family that is closed under union, respectively under union and intersection. It is also proved that $F_k(n)$ is the largest m such that every k-colouring has a monochromatic union-closed family of at least m sets. The following statements are checked.

(1) The bound (1.1), and $F_k(n) \le F(n)$ for $k \ge 2$.
(2) Theorem 1.1 in the stated form $n^{t+1} \le k^{2^t+1} t^t (4M^2)^{t+1} F_k(n)$ for $n \ge 1$, and the bound $F(n) \ge n^t$ for $n \ge 2^{(t+2)^2 2^t}$, together with Lemmas 2.2–4.2 and Propositions 2.5 and 3.2 in the form in which the proof uses them.
(3) Lemma 5.1 and Theorem 1.2, both the bound $F(n) \le \sum_{j<d} \binom{n}{j}$ and the two explicit forms, with the Sauer–Shelah lemma taken from Mathlib.
(4) Corollary 1.3: part (i) with the stated $\omega(n)$ and $\omega(n) \to \infty$, parts (ii)–(iv), and the conjunction of the two conjectures in the form (1.2) as a single theorem.
(5) Proposition 5.2, Lemma 5.3, Propositions 5.4 and 5.5, and Corollary 5.6, with $U(n, N)$ defined literally as the number of union-closed families of N subsets of $[n]$ and the negative answer to Question 9.2 stated for every real constant C.
(6) Corollary 6.1, Theorem 1.5, Proposition 6.2, Theorem 6.3, derived from Mathlib's Hales–Jewett theorem as described above, and Theorem 6.4, proved for every number W with the property that defines $W_k(p)$.
(7) Lemmas 7.1 and 7.2, Proposition 7.5, Lemmas 7.6, 7.7 and 7.8, and Theorem 1.4, including $f(n) > \lceil (n+1)/2 \rceil$ for odd $n \ge 11$ and for $n \ge 23$.

The formal proofs follow the arguments given here, with one change of language: probabilities are replaced by counts. Permutations and size vectors are counted rather than sampled, so that Lemma 4.2 becomes a double counting over the product of the two sets, and the random colourings of Sections 5 and 7 are replaced by counts of colourings. The box inequality is proved for sums over all pairs of points, which is Lemma 3.1 multiplied by $|G|^{2t}$.

The one place where the formal proof takes a different route is the case of Lemma 7.1 in which the colour of a set depends only on its size. Instead of Lemmas 7.3 and 7.4, the formal proof uses a list of 294 explicit sublattices of $2^{[11]}$, each stored as a list of subsets of [11] written in binary, together with the sizes of its members. Two finite statements are checked by evaluation inside the kernel of Lean: that every listed family is closed under union and intersection and has the stated sizes, and that for each of the 2^{11} words g with $g(0)$ blue some listed family has all its sizes in one colour class of g and more members than that class. The second colour is handled by exchanging the two colours. Such a check is a proof in the logic of Lean, verified by

the same kernel as every other proof, and it uses no compiled code and no additional axiom. Lemmas 7.3 and 7.4 themselves are not formalised.

Outside the formal development remain only the figures and the elementary asymptotic rewritings of the proved bounds, such as $\omega(n) = (1-o(1))\log_2\log_2 n$ in Corollary 1.3 and $n^{\log_2\log_2 n-2-o(1)}$ after Proposition 5.2.

9. Questions

With both conjectures settled, what remains open is the first request of Problem 1183: the precise size of the two functions, above all of F.

Question 9.1. *What is the order of magnitude of* $\log F(n)/\log n$*? By Corollary 1.3 it lies between* $(1-o(1))\log_2\log_2 n$ *and* $(1+o(1))\log_2 n$*. Is* $F(n) = n^{\Theta(\log\log n)}$*, or is* $F(n) = n^{\Theta(\log n)}$*?*

Every upper bound for $F(n)$ in this paper comes from the random colouring, and by Proposition 5.2 that colouring cannot give anything below $n^{(1-o(1))\log_2\log_2 n}$. Proposition 5.4 would reduce the matching upper bound to a count.

Question 9.2. *Is there a constant* C *such that* $U(n,N) < 2^{N-1}$ *for all large* n *and all* $N \geq n^{C\log_2\log_2 n}$*?*

A positive answer would have given $F(n) < n^{C\log_2\log_2 n}$ by Proposition 5.4, and together with Corollary 1.3(i) it would have answered Question 9.1 in favour of $n^{\Theta(\log\log n)}$. The answer is negative ([Tr26], Corollary 5.6): by Proposition 5.5, $U(n,N) \geq 2^{N-1}$ for every $N \leq 2^{n/4}$ once $n \geq 206$. Question 9.1 therefore needs an argument that uses more than the number of union-closed families.

Question 9.3. *Is* $f(n) = O(n)$*? If so, does* $f(n)/n$ *converge, and to what limit?*

By Theorem 1.4, $f(n)$ exceeds the chain bound $\lceil (n+1)/2\rceil$ for all large n, and $13/24 \leq f(n)/n \leq (2+o(1))\log_2 n$. The lower bound comes from chains in which stretches of twelve levels are replaced by small sublattices, and the upper bounds from counting sublattices in a random colouring.

Data availability

The Lean formalisation of the results of this paper and the sources of the paper are available at https://doi.org/10.5281/zenodo.23188241. Claude (Anthropic) assisted with the coding.

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